Words of Concern

Point to be As Allan Bloom has said "Education is the movement from darkness to light". Through this handbook, I have tried to illuminate what might otherwise appear as black boxes to some. In doing so, I have used references from several other authors to synthesize or simplify or elaborate information. This is not possible without omitting details that I deem trivial while dilating the data that I consider relevant to topic. Every effort has been made to avoid errors. In spite of this, some errors might have crept in. Any errors or discrepancies noted maybe brought to my notice which I shall rectify in my next revision.

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Thanks,
Karthik v karanam
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Davangere



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Syllabus:
Course Code: SEC-4
Course Title: Cyber Security
Credits - 02
Sem - V
Hours- 30 Theory
Formative Assessment Marks: 25
Summative Assessment Marks: 25
Duration of ESA: 01 hrs.

Course Outcomes
After the successful completion of the course, the student will be able to
1. After completion of this course, students would be able to understand the concept of Cyber security and issues and challenges associated with it.
2. Students, at the end of this course, should be able to understand the cybercrimes, their nature, legal remedies and as to how report the crimes through available platforms and procedures.
3. On completion of this course, students should be able to appreciate various privacy and security concerns on online Social media and understand the reporting procedure of inappropriate content, underlying legal aspects, and best practices for the use of Social media platforms.

Unit Content Duration

Unit 1
Introduction to Cyber Security
Defining Cyberspace and Overview of Computer and Web-technology
Architecture of Cyberspace
Communication and web technology
Internet, World wide web, Advent of internet
Internet infrastructure for data transfer and governance
Internet Society, Regulation of Cyberspace
Concept of cyber security
Issues and challenges of cyber security 9 Hrs

Unit 2:
Cybercrime and Cyberlaw
Classification of cyber crimes
Common cybercrimes-cybercrime targeting computers and mobiles
Cybercrime against women and children
Financial frauds
Social engineering attacks
Malware and ransomware attacks
Zero-day and zero-click attacks
Cybercriminals modus-operandi
Reporting of cyber crimes
Remedial and mitigation measures
Legal perspective of cybercrime

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IT Act 2000 and its amendments
Cybercrime and offences
Organizations dealing with Cybercrime and Cyber security in India.
Case studies. 9 Hrs

Unit 3:
Social Media Overview and Security
Introduction to Social networks
Types of Social media
Social media platforms
Social media monitoring
Hashtag, Viral content
Social media marketing, Social media privacy, Challenges
Opportunities and pitfalls in online social network
Security issues related to social media
Flagging and reporting of inappropriate content
Laws regarding posting of inappropriate content
Best practices for the use of Social media, Case studies. 8 Hrs

Unit 4
Definition of E- Commerce
Main components of E-Commerce
Elements of E-Commerce security
E-Commerce threats
E-Commerce security best practices
Advantage of e-commerce
Survey of popular e-commerce sites
Introduction to digital payments
Components of digital payment and stake holders
Modes of digital payments- Banking Cards
Unified Payment Interface (UPI)
E-Wallets
Unstructured Supplementary Service Data (USSD)
Aadhar enabled payments
Digital payments related common frauds and preventive measures
RBI guidelines on digital payments and customer protection in unauthorized banking transactions Relevant provisions of Payment Settlement Act, 2007 8 Hrs

Unit 5
End Point device and Mobile phone security
Password policy
Security patch management
Data backup
Downloading and management of third-party software
Device security policy
Cyber Security best practices
Significance of host firewall and Ant-virus
Management of host firewall and Anti-virus

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Wi-Fi security
Configuration of basic security policy and permissions 11 Hrs

45 Hrs

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Unit I:

INTRODUCTION TO CYBER SECURITY

Definition Cybersecurity

Cybersecurity experts make sure that hackers can't break in, sensitive information stays private, and everything works smoothly without disruptions from viruses or other online threats.

Cyber security is the practice of defending computers, servers, mobile devices, electronic systems, networks, and data from malicious attacks. It's also known as information technology security or electronic information security.

DEFINING CYBERSPACE AND OVERVIEW OF COMPUTER AND WEB TECHNOLOGY

Cyber Space

Cyberspace refers to the virtual environment created by interconnected digital networks, including the internet, telecommunications networks, computer systems, and other digital communication systems. It encompasses all online interactions, data exchanges, and digital communications, providing a space where information is created, shared, and stored.

Definition: "Cyberspace is a global domain within the information environment consisting of the interdependent network of information systems infrastructures including the Internet, telecommunications networks, computer systems, and embedded processors and controllers."

Overview of Computer

Computer definition:- A computer is a machine that can be programmed to carry out sequences of arithmetic or logical operations (computation) automatically.

A computer is a combination of hardware and software resources that integrate together and provide various functionalities to the user. Hardware is the physical components of a computer like a processor, memory devices, monitor, keyboard, etc., while software is a set of programs or instructions that are required by the hardware resources to function properly.

Components of a Computer

1. Input Unit
2. Central processing unit(CPU)
3. Output Unit

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1. Input Unit

The input unit consists of devices that are attached to the computer. These devices take input and convert it into binary language that the computer understands. Some of the common input devices are keyboard, mouse, joystick, scanner etc.
The input unit is formed by attaching one or more input devices to a computer
A user input data and instructions through input devices such as a keyboard, mouse, etc.
The input unit is used to provide data to the processor for further processing

2. Central Processing unit

The Central Processing Unit (CPU) is responsible for executing instructions and performing calculations in a computer system. It processes data and controls the overall operation of the computer, often referred to as the "brain" of the system.

The CPU has two main components: the Control Unit (CU) and the Arithmetic Logic Unit (ALU). The CU manages instruction execution by fetching and decoding instructions, while the ALU performs arithmetic and logical operations as instructed by the program.

3. Output unit

The output unit is a collection of devices that receive binary data from the CPU and convert it into a human-readable format. Common output devices include monitors, printers, and plotters.

Computers have several key characteristics:

Speed, accuracy, diligence, versatility, and storage. They can perform millions of calculations per second with extreme accuracy, complete complex and lengthy calculations simultaneously, and carry out multiple operations at once. Additionally, computers are designed to store and retrieve large amounts of data and instructions at any time.

Advantages of the computer.

- High speed.
- Accuracy.
- Storage capability.
- Diligence.
- Versatility.
- Reliability.
- Automation

Disadvantages of computer

- Dependency
- Environment
- No feeling

OVERVIEW OF WEB-TECHNOLOGY

Web Technology refers to the various tools and techniques that are utilized in the process of communication between different types of devices over the Internet. A web browser is used to access web pages. Web browsers can be defined as programs that display text, data, pictures, animation, and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers

Web Technology can be classified into the following sections:

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- **World Wide Web (WWW):** The World Wide Web is based on several different technologies: Web browsers, Hypertext Markup Language (HTML), and Hypertext Transfer Protocol (HTTP).
- **Web Browser:** The web browser is an application software to explore www (World Wide Web). It provides an interface between the server and the client and requests to the server for web documents and services.
- **Web Server:** Web server is a program which processes the network requests of the users and serves them with files that create web pages. This exchange takes place using Hypertext Transfer Protocol (HTTP).
- **Web Pages:** A webpage is a digital document that is linked to the World Wide Web and viewable by anyone connected to the internet has a web browser.
- **Web Development:** Web development refers to the building, creating, and maintaining of websites. It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet i.e. Websites.

Web Development can be classified into two ways:

- **Frontend Development:** The part of a website that the user interacts directly is termed as front end. It is also referred to as the 'client side' of the application.
- **Backend Development:** The backend is the server side of a website. It is the part of the website that users cannot see and interact. It is the portion of software that does not come in direct contact with the users. It is used to store and arrange data.
- **Frontend Languages:** The front-end portion is built by using some languages which are discussed below:
- **HTML:** HTML stands for Hypertext Markup Language. It is used to design the front-end portion of web pages using a markup language. HTML is a combination of Hypertext and Markup language. Hypertext defines the link between the web pages. The markup language is used to define the text documentation within the tag which defines the structure of web pages.
- **CSS:** Cascading Style Sheets fondly referred to as CSS is a simply designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page.
- **Javascript:** javascript is a famous scripting language used to create magic on the sites to make the site interactive for the user. It is used to enhancing the functionality of a website to run cool games and web-based software.
- **AJAX:** Ajax is an acronym for Asynchronous Javascript and XML. It is used to communicate with the server without refreshing the web page and thus increasing the user experience and better performance.

Backend Languages: The back end portion is built by using some languages which are discussed below:

- **PHP:** PHP is a server-side scripting language designed specifically for web development. Since PHP code executed on the server-side, so it is called a server-side scripting language.
- **Node.js:** Node.js is an open-source and cross-platform runtime environment for executing javascript code outside a browser. You need to remember that nodejs is not a framework, and it's not a programming language. Most people are confused and understand it's a framework or a programming language. We often use Node.js for building back-end services like apis like Web App or Mobile App. It's used in production by large companies such as Paypal, Uber, Netflix, Wallmart, and so on.
- **Python:** Python is a programming language that lets you work quickly and integrate systems more efficiently.
- **Ruby:** Ruby is a dynamic, reflective, object-oriented, general-purpose programming language. Ruby is a pure Object-Oriented language developed by Yukihiro Matsumoto. Everything in Ruby is an object except the blocks but there are replacements too for it i.e procs and lambda. The objective of Ruby's development was to make it act as a sensible buffer between human programmers and the underlying computing machinery.
- **Java:** Java is one of the most popular and widely used programming languages and platforms. It is highly scalable. Java components are easily available.
- **Javascript:** javascript can be used as both (front end and back end) programming.

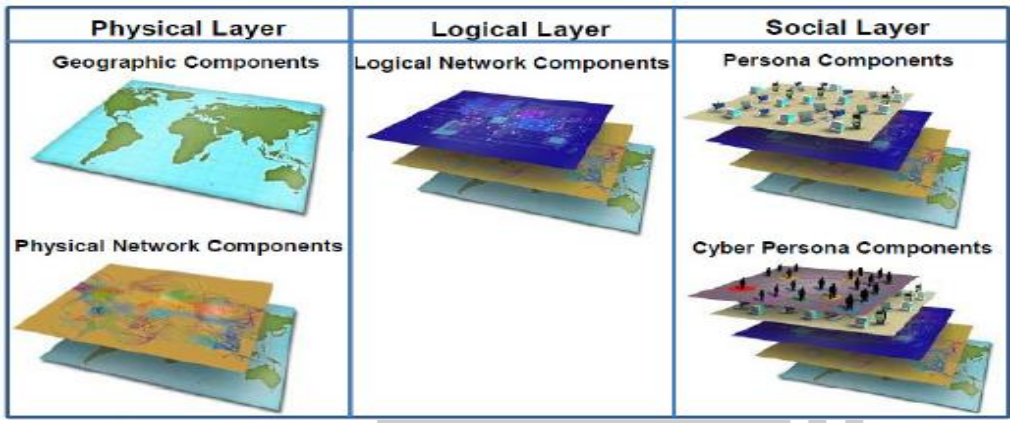
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- **Golang:** Golang is a procedural and statically typed programming language having a syntax similar to C programming language. Sometimes it is termed as Go Programming Language.
- **C#:** C# is a general-purpose, modern, and object-oriented programming language pronounced as “C sharp”.
- **DBMS:** The software which is used to manage a database is called Database Management System (DBMS).

ARCHITECTURE OF CYBERSPACE

Definition: Cyberspace refers. To the interconnected digital environment created by computer systems and networks. While it doesn't have a physical presence, it's a conceptual space where digital mt~ract10ns, communications, and transactions occur. The architecture of cyberspace involves various components and layers that enable its functionality

Cyberspace can be viewed as three layers (physical, logical, and social) made up of five components (geographic, physical network, logical network, cyber persona, and persona).



Physical layer

The physical layer includes the geographic component and the physical network component. The geographic component is the physical location of elements of the network. While geopolitical boundaries can easily be crossed in cyberspace at a rate approaching the speed of light, there is still a physical aspect tied to the other domains. At the lowest level, cyberspace is supported by physical infrastructure, including data centers, servers, routers, switches, and undersea fiber optic cables. These components facilitate the transmission of data and signals across the globe.

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Logical layer

The logical layer contains the logical network component, which is technical and consists of the logical connections that exist between network nodes. Nodes are any devices connected to a computer network. Nodes can be computers, personal digital assistants, cell phones, or various other network appliances. On an Internet protocol (IP) network, a node is any device with an IP address.

Social layer

The social layer comprises the human and cognitive aspects, including the cyber persona and persona components. The cyber persona component includes a person’s identification

or persona on the network (e-mail address, computer IP address, cell phone number, and others). The persona component consists of the people actually on the network. An individual can have multiple cyber personas (for example, different e-mail accounts on different computers) and a single cyber persona can have multiple users.

Network Layer: The network layer forms the backbone of cyberspace. It consists of various interconnected networks, such as local area networks (LANs), wide area networks (WANs), and the Internet itself. This layer provides the means for devices to communicate with each other over long distances.

Protocols and Standards: Protocols and standards define the rules for how data is transmitted, received, and processed in cyberspace. Examples include TCP/IP (Transmission Control Protocol/Internet Protocol), HTTP (Hypertext Transfer Protocol), and DNS (Domain Name System). These protocols ensure that devices and systems can understand and communicate with each other effectively.

Software Layer: The software layer encompasses the various applications, services, and platforms that users interact with in cyberspace. This includes web browsers, email clients, social media platforms, online marketplaces, and more. Software developers create applications that leverage the underlying infrastructure to provide specific functionalities and services.

User Interfaces: User interfaces (Uls) are the points of interaction between users and the digital environment. These interfaces can range from graphical user interfaces (GUIs) in applications to command-line interfaces (CLis) for more technical interactions. The design of user interfaces greatly influences the user experience within cyberspace.

Cyber security Measures: Given the interconnected nature of cyberspace, cyber security is crucial. Security measures such as firewalls, encryption, authentication, and intrusion detection systems are integral components of the architecture to protect against cyber threats and attacks.

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Virtual Spaces and Environments: In addition to the practical infrastructure and communication tools, there are virtual spaces and environments within cyberspace. These include online multiplayer games, virtual reality platforms, and digital simulations where users can interact with each other and the environment.

Decentralization and Blockchain: With the rise of technologies like blockchain, cyberspace architecture has begun to incorporate decentralized systems. Blockchain allows for secure and transparent transactions without the need for a central authority. This is particularly relevant in areas like cryptocurrencies and digital identity management.

COMMUNICATION AND WEB TECHNOLOGY

Communication and web technology play a crucial role in shaping how people connect, collaborate, and share information in the digital age. Technology has transformed the way we communicate and have given rise to the modern internet-driven world. The term "web technology" refers to a variety of tools and methods used in internet-based communication between various sorts of devices. To access web pages, utilize a web browser. Programs known as web browsers are those that show text, data, images, animations, and videos on the internet. Web browser software interfaces can be used to access world wide web resources that are hyperlinked.

Communication

Communication over the internet is the process of exchanging information between two or more people using electronic devices and the Internet. It can be done through a variety of channels, such as email, instant messaging, social media, and video conferencing.

Cyber security involves the protection of digital systems, networks, and data from various forms of cyber threats
Data Transmission
Encryption
Confidentiality
Authentication and Identity

Data Transmission: Data transmission refers to the process of sending digital information, such as text, images, videos, or any type of electronic data, from one point to another. This process involves the movement of data through various channels, mediums, or networks, enabling communication and information exchange between devices, systems, or individuals

Encryption: Encryption is a process of converting readable or plaintext data into a scrambled form called ciphertext using algorithms and keys. This technique ensures data confidentiality and security by making the information unreadable to unauthorized individuals who don't possess the appropriate decryption key

Confidentiality: Confidentiality is one of the key principles of information security, focusing on ensuring that sensitive and private data is accessible only to those individuals who have the authorized permission to access it. In the context of cybersecurity, maintaining confidentiality is crucial for protecting sensitive information from unauthorized disclosure or access.

Authentication and Identity: Authentication and identity are crucial aspects of communication in the context of cybersecurity. They play a vital role in ensuring that the participants in a communication exchange are who they claim to be, thus establishing trust and protecting sensitive information.

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Communication Technologies

Communication technologies facilitate the exchange of information between individuals between individuals and entities. In the context of the web communication technologies include

1. Email: Electronic mail is one of the earliest forms of online communication, allowing users to send text, files, and multimedia content to recipients.

2. Instant Messaging (IM): Services like WhatsApp, Facebook Messenger, and Slack enable real-time text-based communication.

3. Voice over IP (VoIP): Technologies like Skype and Zoom use the internet to transmit voice and video calls.

4. Social Media: Platforms like Facebook, Twitter, Instagram, and LinkedIn provide spaces for users to share content, connect with others, and engage in discussions.

5. Video Sharing and Streaming: Platforms like YouTube, Vimeo, and Twitch allow users to upload and view videos, as well as live stream content.

6. Virtual Reality (VR) and Augmented Reality (AR): These technologies offer immersive communication experiences, allowing users to interact in virtual or augmented environments.

Web Communication Tools

Web communication tools leverage web technologies to facilitate various forms of interaction:

1. Blogs and Content Management Systems (CMS): Tools like WordPress, Joomla, and Drupal enable users to create and manage websites, blogs, and digital content. .

2. Forums and Discussion Boards: Online communities use platforms like Redd1t, Stack Exchange, and phpBB for discussions on various topics.

3. Collaboration Tools: Services like Google Workspace (formerly G Suite), Microsoft365, and Slack enhance teamwork through document sharing, real-time ed1tmg, and communication channels.

WORLD WIDE WEB

- The World Wide Web, often simply referred to as the "web," is a global system of interconnected documents and resources that are linked together by hyperlinks and accessed via the Internet. It is a vast network of information, websites, multimedia content, and applications that users can access using web browsers.
- The World Wide Web was invented in 1989 by Tim Berners-Lee at CERN, a European research organization. Berners-Lee envisioned a system where scientists could share information more easily. He created the first web server and wrote the first web browser, which was called World Wide Web.
- The World Wide Web is based on a client-server model. This means that web servers store web pages and web browsers retrieve them. Web browsers are software applications that allow users to view web pages.
- The World Wide Web is powered by the Internet. The Internet is a global network of computers that are connected to each other. The Internet allows web servers and web browsers to communicate with each other.

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Here are some of the key technologies that have made the World Wide Web possible:

Hypertext Transfer Protocol (HTTP): HTTP is the protocol that is used to transfer web pages between web servers and web browsers.
HTML (hypertext Markup Language): HTML is the markup language that is used to create web pages.
CSS (Cascading Style Sheets): CSS is used to style web pages.
Javascript: javascript is a programming language that is used to add interactivity to web pages.
Web servers: Web servers are computers that store web pages and make them available to web browsers.
Web browsers: Web browsers are software applications that allow users to view web pages.

Here are key points about the World Wide Web:

Web Browsers: Web browsers are software applications that allow users to access and view web content. Popular web browsers include Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.

Uniform Resource Locator (URL): A URL is a web address that specifies the location of a specific web page or resource on the internet. Urls consist of the protocol (e.g., http, https), domain name, and path.
Web Servers and Hosting: Web servers store and serve web content to users' browsers. Websites are hosted on web servers and made accessible to users via the internet.

HTTP and HTTPS: hypertext Transfer Protocol (HTTP) is the foundation of data communication on the web. HTTPS adds an extra layer of security by encrypting the data transmitted between users and websites.

Web Content Types: The web hosts a wide variety of content, including text, images, videos, audio, interactive applications, and more.

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Web Standards: Organizations like the World Wide Web Consortium (W3C) establish standards and guidelines to ensure interoperability and accessibility of web technologies.

Search Engines: Search engines like Google, Bing, and Yahoo enable users to search for specific information across the vast expanse of the web.

Web 2.0 and Beyond: Web 2.0 introduced user-generated content, social networking, and interactive web applications. The term "Web 3.0" refers to the vision of a more intelligent, semantic web with increased interconnectivity and data integration.

E-Commerce and Online Services: The web has enabled e-commerce, allowing businesses to sell products and services online. It also hosts various online services like email, social media, streaming, and more.

Responsive Web Design: Designing websites to adapt to various screen sizes and devices ensures a consistent user experience across platforms.

Web Accessibility: Web accessibility ensures that websites are usable by people with disabilities, promoting inclusivity and equal access to information.

Global Reach and Information Exchange: The web enables instant access to information from around the world, facilitating communication, education, and collaboration on a global scale.

Challenges and Concerns: The web has brought about challenges such as misinformation, online privacy concerns, cybersecurity threats, and the digital divide.

ADVENT OF INTERNET

The Internet has revolutionized the way we live, work, and communicate. It has made it possible for us to connect with people from all over the world, access information instantly, and do business online.
The advent of the Internet can be traced back to the early 1960s when the United States Department of Defense began developing a network of interconnected computers. This network, called ARPANET(1969), was designed to be resilient to attack and to allow researchers to share information quickly and easily.
In the 1970s, ARPANET began to be used by universities and other institutions around the world. In 1983, the TCP/IP protocol was developed, which standardized the way that computers communicate over the network. This led to the development of the World Wide Web in the early 1990s, which made it possible for anyone with a computer to access information and resources on the Internet.

The World Wide Web: A Game-Changer
While ARPANET laid the foundation for the Internet, it was the invention of the World Wide Web (WWW) by British computer scientist Tim Berners-Lee in 1989 that truly democratized information access. The WWW introduced the concept of hyperlinks, allowing users to navigate easily between web pages. It made the internet user-friendly and accessible to the masses.

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The Internet has had a profound impact on society. It has made it possible for people to communicate and collaborate more easily, and it has opened up new opportunities for businesses and individuals. The Internet has also been a powerful tool for social change, as it has allowed people to connect and organize around shared causes.
Impact of the Internet

The Internet has had a significant impact on communication, education, business, and society as a whole.

Communication: The Internet has made it possible for people to communicate with each other more easily than ever before. We can now use email, social media, and video chat to stay in touch with friends and family members all over the world. The Internet has also made

it possible for people to connect with others who share their interests, regardless of their location.

Education: The Internet has revolutionized education. Students can now access educational resources from all over the world, and they can learn at their own pace. The Internet has also made it possible for students to collaborate with each other and with their teachers on projects.

Business: The Internet has transformed the way that businesses operate. Businesses can now reach customers all over the world, and they can sell their products and services online. The Internet has also made it possible for businesses to collaborate with each other more easily.

Society: The Internet has had a significant impact on society as a whole. It has made it possible for people to connect with each other and to share information more easily. The Internet has also been a powerful tool for social change, as it has allowed people to connect and organize around shared causes.

The Social Media Revolution

The 21st century brought with it the advent of social media platforms like Facebook, Twitter, and Instagram. These platforms revolutionized the way we connect with friends and family, share our lives, and even influence public discourse. Social media has been a powerful tool for social and political movements, enabling global connectivity and the rapid spread of ideas.

The Internet of Things (iot)

In recent years, the internet has expanded beyond computers and smartphones to include everyday objects. The Internet of Things (iot) connects devices like refrigerators, thermostats, and cars to the Internet, allowing them to communicate and perform tasks autonomously. This technological shift is reshaping industries, from healthcare to transportation

The Internet is a powerful tool that has had a profound impact on society. It has made it possible for us to connect with people from all over the world, access information instantly, and do business online. However, the Internet also poses some challenges, such as the spread of misinformation, crime, and addiction. It is important to be aware of these challenges and to take steps to mitigate them.

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INTERNET INFRASTRUCTURE FOR DATA TRANSFER AND GOVERNANCE

The internet infrastructure is the physical and logical components that enable the Internet to function. It is a complex and interconnected system that includes:

The physical infrastructure: This includes the cables, wires, and other physical components that carry data over the internet. It also includes the telecommunications towers and data centers that house the equipment that powers the internet.

The logical infrastructure: This includes the protocols, standards, and software that enable devices to communicate with each other over the internet. It also includes the domain name system (DNS), which translates human-readable web addresses into machine-readable addresses that computers use to communicate with each other.

The Internet infrastructure is owned and operated by a variety of organizations, including:

Internet service providers (isps): isps provide the connections that allow users to access the Internet.
Content delivery networks (cdns): cdns deliver content, such as web pages and videos, to users more quickly by caching it in servers that are located closer to the users.
Internet exchange points (ixps): ixps are points where isps exchange traffic with each other.
Research and education networks (rens): rens provide high-speed internet connectivity to research and educational institutions

DATA TRANSFER

Data transfer is the process of moving data from one location to another. This can be done over a variety of networks, including wired networks, wireless networks, and the Internet.

Data is transferred over the internet using a technique called packet switching. In packet switching, data is broken up into small chunks called packets. Each packet contains information about where it came from, where it needs to go, and how many packets make up the entire message. The packets are then routed through the internet infrastructure, which is a network of interconnected networks.

The internet infrastructure is made up of a variety of devices, including:

- **Routers:** Routers are devices that direct packets through the internet. They do this by looking at the destination address in each packet and routing it to the next hop on the way to its destination.
- **Switches:** Switches are devices that connect devices on the same network. They do this by looking at the MAC address of each device and routing the packet to the correct port.
- **Servers:** Servers are devices that store data and provide services to other devices on the internet.
- **Modems:** Modems are devices that convert digital signals to analog signals and vice versa. This allows devices to communicate over different types of networks, such as telephone lines and cable lines.

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When you send a message over the internet, your computer breaks the message up into packets and sends them to the router that is connected to your network. The router then routes the packets to the next hop on the way to their destination. This process continues until the packets reach the server that is hosting the website or service that you are trying to access. The server then reassembles the packets and sends the message back to your computer.

The speed at which data is transferred over the internet depends on a number of factors, including:

- **The bandwidth of the network:** The bandwidth is the amount of data that can be transferred over a network in a given amount of time.
- **The distance between the sender and receiver**: The further the distance, the longer it takes for the packets to travel.
- **The number of routers and switches that the packets have to pass through**: The more routers and switches, the longer it takes for the packets to reach their destination.
- **The amount of traffic on the network:** The more traffic on the network, the slower the data transfer speeds.

DATA GOVERNANCE

Data governance is the set of policies, processes, and standards that ensure that data is managed effectively and efficiently throughout its lifecycle. This includes data collection, storage, processing, use, and disposal.

Internet infrastructure is essential for data governance because it is the foundation on which data is collected, stored, processed, and shared. The internet infrastructure provides the channels through which data can flow, and it also provides the tools and technologies that can be used to manage data.

Here are some of the ways in which Internet infrastructure can be used for data governance:

Data collection: The internet infrastructure can be used to collect data from a variety of sources, including websites, sensors, and social media platforms. This data can then be used to improve decision-making, identify trends, and develop new products and services.

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Data storage: The internet infrastructure can be used to store data in a secure and reliable manner. This data can then be accessed and used by authorized users.

Data processing: The internet infrastructure can be used to process data in a variety of ways, such as by analyzing data to identify patterns and trends. This data can then be used to improve decision-making and make better business decisions.

Data sharing: The internet infrastructure can be used to share data with authorized users. This data can then be used to collaborate on projects, develop new products and services, and improve decision-making.

INTERNET SOCIETY

The Internet Society (ISOC) is an American non-profit advocacy organization founded in 1992 with local chapters around the world. Its mission is "to promote the open development, evolution, and use of the Internet for the benefit of all people throughout the world."

The Internet Society (ISOC) is a non-profit organization that promotes the open development, evolution, and use of the Internet for the benefit of all. ISOC was founded in 1992 by a group of Internet pioneers, including Vinton Cerf, Robert Kahn, and Tim Berners-Lee.

- ISOC is a member-driven organization with over 90,000 members in over 180 countries. ISOC's members include individuals, organizations, and companies that are involved in the Internet community.
- ISOC is a leading voice for the open Internet. ISOC works to ensure that the Internet remains a global public resource that is accessible, affordable, and open to all.

The Internet Society has a number of goals, including:

- Ensuring that the Internet remains open and accessible to everyone
- Promoting the development of new Internet technologies and applications
- Educating the public about the Internet and its benefits
- Advocating for policies that support the open and secure use of the Internet

What Does the Internet Society Do?

- **Advocacy:** ISOC speaks up for the internet. Just like you have someone to stand up for you when you need help, ISOC stands up for the internet and its users. It talks to governments, businesses, and other important people to make sure they understand how to keep the Internet working well.
- **Making Rules:** Imagine a game without any rules; it would be chaotic. The internet has rules too, called standards. ISOC helps make these standards. They ensure that when you send an email or visit a website, everything works smoothly.
- **Helping People Learn:** ISOC is like a teacher for the internet. It helps people learn about the internet and how to use it safely. This is super important because the internet can sometimes be a tricky place, and we all need to know how to stay safe.
- **Connecting the World:** Not everyone has access to the internet, which is sad because it's such a great tool for learning and staying in touch. ISOC works to connect more people to the internet, especially in places where it's hard to get online.
- **Keeping Us Safe:** Just like your parents want you to be safe when you play outside, ISOC wants to keep you safe on the internet. It helps create rules and tools to protect your privacy and security.

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The Internet Society is important because it makes sure the Internet remains a friendly and useful place for everyone. Here's why it matters:

- **Equal Access:** ISOC wants everyone to have a chance to use the internet, no matter where they live or how much money they have. This helps make the world more fair.
- **Safety:** By setting rules and helping people stay safe online, ISOC makes sure we can enjoy the internet without worrying too much about bad things happening.
- **Learning:** ISOC helps us learn and share knowledge on the internet. It's like a giant library and classroom that's always open!
- **Innovation:** Without ISOC, the internet might not work as well as it does today. It helps keep the internet exciting and full of new ideas.

REGULATION OF CYBERSPACE

Cyberspace refers to the virtual computer world, and more specifically, is an electronic medium used to form a global computer network to facilitate online communication. It is a large computer network made up of many worldwide computer networks that employ TCP/IP protocol to aid in communication and data exchange activities. Cyberspace's core feature is an interactive and virtual environment for a broad range of participants.

The regulation of cyberspace is a complex and evolving challenge. Cyberspace is a global space, and it is difficult to regulate it using traditional national laws and regulations. Additionally, the technology of cyberspace is constantly changing, which makes it difficult to develop regulations that can keep up with the pace of innovation.

In India, the Information Technology Act, of 2000 is the legislation that covers the domain of cyber law. The main objective of the Act is to provide legal recognition for transactions carried out using electronic data interchange and other means of electronic communication, commonly referred to as e-commerce, which involve the use of alternatives to paper-based methods of communication and storage of information to facilitate electronic filing of documents with the Government agencies.

Some of the key areas that need to be regulated in cyberspace include:

- **Cybersecurity:** This includes measures to protect against cyberattacks, such as malware, phishing, and data breaches.
- **Privacy:** This includes protecting personal information that is collected and stored online.
- **Intellectual property:** This includes protecting copyrights, trademarks, and patents that are used online.
- **Child safety:** This includes protecting children from harmful content and online predators.
- **Hate speech:** This includes preventing the spread of hate speech and other forms of online abuse.

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Government policies and laws regarding the regulation of Internet content

According to Electronic Frontiers Australia (EFA, March 2002) report on government policies regarding internet censorship in various countries, government policies can be classified into the following four categories:
1) The policy to encourage self-regulation.
2) Criminal law penalties (Fines or Jail Terms) applicable to content providers who make content “unsuitable for minors” available online.
3) The government has also mandated blocking of access to content deemed unsuitable for adults; for example: Australia, China, Saudi Arabia, Singapore etc. Page 7 of 18
4) A number of countries have either prohibited general public access to the internet or require internet users to be registered/licensed by a government authority before permitting them restricted access.

Here are some of the ways that cyberspace can be regulated

- **Government regulation**: Governments can enact laws that specifically address cybercrime and other online threats. Additionally, governments can develop regulations that promote cybersecurity and protect the privacy of Internet users.
- **Self-regulation:** The private sector can also play a role in regulating cyberspace. For example, companies can develop their own cybersecurity standards and privacy policies. Additionally, companies can work together to address online threats such as spam and malware.
- **International cooperation:** Governments and the private sector can also work together to regulate cyberspace on an international scale. This can be done through organizations such as the G7 and the Internet Society.

CONCEPT OF CYBER SECURITY

Cyber security is the practice of protecting systems, networks, and data from digital attacks. It is a broad term that encompasses a wide range of security measures, from technical solutions to security policies and procedures.

The concept of cyber security is based on the three pillars of confidentiality, integrity, and availability (also known as the CIA triad).
Confidentiality refers to the protection of data from unauthorized access.
Integrity refers to the protection of data from unauthorized modification.
Availability refers to the protection of data from unauthorized destruction or disruption.

Cybersecurity is important for businesses, organizations, and individuals. Businesses need to protect their data from cyberattacks to prevent financial losses, reputational damage, and disruption to operations. Organizations need to protect their data from cyberattacks to protect the privacy of their users and to prevent the misuse of their data. Individuals need to protect their data from cyberattacks to protect their financial information, personal information, and intellectual property.

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Types of cybersecurity threats

Phishing
Phishing is the practice of sending fraudulent emails that resemble emails from reputable sources. The aim is to steal sensitive data like credit card numbers and login information. It's the most common type of cyberattack. You can help protect yourself through education or a technology solution that filters malicious emails.

Social engineering
Social engineering is a tactic that adversaries use to trick you into revealing sensitive information. They can solicit a monetary payment or gain access to your confidential data. Social engineering can be combined with any of the threats listed above to make you more likely to click on links, download malware, or trust a malicious source.

Ransomware
Ransomware is a type of malicious software. It is designed to extort money by blocking access to files or the computer system until the ransom is paid. Paying the ransom does not guarantee that the files will be recovered or the system restored.

Malware
Malware is a type of software designed to gain unauthorized access or to cause damage to a computer.

There are many different types of cyberattacks, including:

Data breaches: A data breach is an incident in which sensitive data is exposed to unauthorized individuals. This can happen through hacking, human error, or other means.

SQL Injection Attack: Structured Query Language (SQL) injection is a common method of taking advantage of websites that depend on databases to serve their users. Clients are computers that get information from servers, and an SQL attack uses an SQL query sent from the client to a database on the server. The command is inserted, or "injected", into a data plane in place of something else that normally goes there, such as a password or login. The server that holds the database then runs the command and the system is penetrated.

DDoS (Distributed denial-of-service) attacks: A DDoS attack is an attempt to overwhelm a website or server with traffic so that it becomes unavailable.

DNS Spoofing: With Domain Name System (DNS) spoofing, a hacker alters DNS records to send traffic to a fake or "spoofed" website. Once on the fraudulent site, the victim may enter sensitive information that can be used or sold by the hacker. The hacker may also construct a poor-quality site with derogatory or inflammatory content to make a competitor company look bad.

Zero-day attacks: A zero-day attack is an attack that exploits a vulnerability in software that the software vendor is not aware of. These attacks are often very difficult to defend against.

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There are a number of different ways to protect cybersecurity. Some of the most common methods include:

Using strong passwords and two-factor authentication: Strong passwords and two-factor authentication can help to prevent unauthorized access to accounts and systems.
Keeping software up to date: Software updates often include security patches that can help to protect against known vulnerabilities.
Using a firewall: A firewall can help to protect networks from unauthorized access.
Using antivirus and anti-malware software: Antivirus and anti-malware software can help to protect systems from malware attacks.
Educating employees about cybersecurity: Employees should be educated about cybersecurity risks and best practices.

ISSUES AND CHALLENGES OF CYBER SECURITY

ISSUES

- **Cyberattacks and threats:** Organizations and individuals are at serious risk due to the increase in cyberattacks, including malware, ransomware, phishing, and advanced persistent threats.
- **Data breaches:** Data breaches are a major cybersecurity concern. When a data breach occurs, sensitive data such as personal information, financial information, or intellectual property can be exposed. Data breaches can have a significant impact on businesses and individuals.
- **Cloud security:** As more and more businesses move their data and applications to the cloud, cloud security is becoming a top priority. Cloud providers offer a variety of security features, but it is important for businesses to understand these features and implement them correctly.
- **Iot Vulnerabilities:** Inadequate security measures in interconnected devices can lead to network breaches.
- **Zero-day attacks:** Zero-day attacks are attacks that exploit vulnerabilities in software that the software vendor is not aware of. Zero-day attacks are often very difficult to defend against, as there is no patch available to fix the vulnerability.
- **Cyberwarfare:** Cyberwarfare is the use of cyberattacks by nation-states or other entities to achieve political or military goals. Cyberwarfare is a growing threat, and it is important for businesses and individuals to be aware of the risks.
- **Skills shortage:** There is a global shortage of cybersecurity skills. This shortage is making it difficult for businesses to protect themselves from cyberattacks.
- **Data Privacy and Compliance:** Stricter data protection regulations, such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA), place greater emphasis on protecting user data and holding organizations accountable for breaches.

To protect yourself from cyber threats, you can take the following steps:

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Be aware of the latest cyber threats.
Keep your software up to date.
Use strong passwords and don't share them with anyone.
Be careful about what links you click on and what attachments you open.
Only download files from trusted sources.
Use a firewall and antivirus software.
Back up your data regularly.
Educate yourself about cybersecurity.

CHALLENGES

The increasing sophistication of cyber threats: Cybercriminals are constantly developing new and more sophisticated ways to attack computer systems and networks. This makes it difficult for businesses and individuals to stay ahead of the curve.

The growing number of connected devices: The Internet of Things (IOT) is rapidly growing, and this is increasing the attack surface for cybercriminals. IOT devices are often not as secure as traditional computers and networks, making them easy targets for attack.

The increasing complexity of cybersecurity regulations: Governments around the world are increasingly imposing regulations on businesses and organizations to improve their cybersecurity posture. This can be a challenge for businesses, as they need to comply with a variety of regulations that may be conflicting or overlapping.

Insider Threats: Employees or partners with access to systems can intentionally or accidentally compromise security.

The growing use of cloud computing: Cloud computing is becoming increasingly popular, but it also introduces new security challenges. Businesses need to ensure that their cloud data is secure and that they are using cloud providers that have strong security practices.

Lack of Cybersecurity Awareness: Users' inadequate understanding of risks makes them susceptible to social engineering tactics.

Third-party risk: Third-party vendors play an increasingly important role in the modern enterprise. However, third-party vendors can also pose a security risk. It is important for businesses to carefully vet their third-party vendors and to monitor their security posture on an ongoing basis.

Mobile Security: The widespread use of mobile devices presents unique challenges for security, as these devices are often used for both personal and work-related tasks and are susceptible to various mobile-specific threats.

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Unit-2

CYBER CRIME AND CYBER LAW

CYBERCRIME

Cybercrime refers to illegal activities carried out using computers, digital devices, or the internet. It involves exploiting technology to commit crimes such as hacking, identity theft, fraud, data breaches, and more. Cybercrimes can target individuals, organizations, or governments, causing financial, emotional, or reputational damage.

The term “cybercrime” refers to any illegal behavior done on or through a computer the internet, or any other technology regulated by the "Information Technology Act."

Cybercrime is defined as "criminal activities carried out via computers or the Internet". "Those species whose genus is traditional crime and where the computer is either an object or a subject of the criminal conduct can be defined as cybercrime."

CYBER LAW

Cyber law, also known as internet law or IT law, refers to the legal regulations and rules that govern activities conducted in cyberspace, including the use of the internet, digital communications, and electronic transactions. It aims to protect individuals, businesses, and governments from cybercrimes, ensure data privacy, and maintain fair usage of technology.

Cyber law, also known as cyber law or internet law, refers to the legal framework that governs and regulates activities in the digital and online space. It encompasses a wide range of legal issues related to the Internet, computers, information technology, and digital communication. Cyberlaw addresses the rights, responsibilities, and obligations of individuals, organizations, and governments in the context of the digital world.

CLASSIFICATION OF CYBER CRIMES

There are three major categories that cybercrime falls into:

1. **Cyber Crime against individual**
2. **Cyber Crime against property**
3. **Cyber Crime against organizations**
4. **Cyber Crimes Against Society**

1. Cyber Crime against individual

Cybercrime against individuals" refers to illegal activities carried out in the digital domain that specifically target individual people or their personal information for various harmful purposes. These criminal acts are committed using computer systems, networks, and the internet, often with the intent to exploit, defraud, harass, or otherwise harm individuals.

- Cyber crimes may affect individuals in different manners like
  E-mail spoofing, spamming, cyber defamation, phishing, cyberstalking.

2. Cyber Crime against property

Cybercrime against property involves damaging or stealing a person's or an organization's property.

- These crimes are usually economic in nature and involve
- Credit card skimming, theft of Intellectual Property, Identity Theft, domain name dispute,
- Software Piracy, E-mail fraud, digital vandalism.

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3. Cyber Crime against Organisations

Cybercrime also affects organizations like banks, service sectors, government agencies, companies, and other associations of persons.

- These involve Hacking, Denial of Service, Viruses, and Worms, E-mail bombings, salami attacks, Logic bombs, Spyware, etc.

4. Cyber Crimes Against Society

Crime committed affects the interest of society at large and against the public, including Cyber terrorism, Cyber attacks on institutions, Fake news, Online gambling.

Types of Cybercrime

1. One of the most serious violations is the use of child pornography: Predators all throughout the world interact with youngsters online and sexually attack them. The growing usage of the internet has made children an appealing target for hackers. In order to lure children into their traps, pedophiles approach them in chat rooms, make friends with them, and steal personal information from their helpless victims. Kids are lured to the internet by these pedophiles, who then attack them sexually or use them as sex objects.

2. Cyberbullying: A cyberbully is someone who uses electronic tools like computers, mobile phones, laptops, etc. To harass or bully others. Bullying committed via digital technology is referred to as cyberbullying. It could involve the usage of social media, messaging services, gaming platforms, and mobile devices. This frequently entails repeating actions meant to frighten, infuriate, or degrade the targets.

3. Hacking: Hacking is the technique of illegally accessing device information, making changes to allow for continued access, and changing the configuration, use, or maintenance of the target equipment without the knowledge or consent of the owners.

4. Cyber-grooming: This phenomenon involves someone becoming close to an adolescent and then using coercion, taunting, or other methods to get them to engage in sexual activity.

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5. Dissemination of virus: This type of illegal activity necessitates direct or unauthorized access to the operating system through the installation of additional software that falls under the headings of ss bugs, worms, or logic bombs. It is obvious that computer sabotage, which generally refers to the improper deletion or destruction of computer data or the Internet feature that interferes with ordinary device functionality, is illegal.

6. Phishing: Since it is simple for modern organizations to trade cash for money stored in computers, this practice results in computer theft. Organized crime usually targets. Personal and financial credit card information as well as credit card identity information. Digital assets may potentially belong to a higher economic class, in addition to having a far higher value than conventional -economic assets.

7. Spoofing: Having one machine on a network have a separate computer, typically a computer with privileged access, enables worldwide access to the other machines.

8. Internet stalking: Cyberstalking, as used in modern culture, is the practise of following or keeping an eye on someone online. A cyber-stalker does not physically pursue their target; instead, they find them after an online interaction and verbally stalk, harass, and threaten them. It violates your right to privacy on the internet.

9. Online job fraud: An online job fraud strategy involves deceiving job seekers by making. Better, higher-paying jobs sound appealing while actually only providing them false hope. The Reserve Bank of India(RBI) issued a warning on March 21, 2022, urging citizens not to be duped by job scams by doing this, the RBI has described how online job fraud is carried out and the safety measures the average person should take while applying for any job, whether it be in India or abroad.

10. Sexting online: When a cybercriminal threatens someone with publishing private and sensitive mformat10n online, this is known as online sextortion. These criminals use threats to coerce victims into providing them with a sexual image, a sexual favor, or money.

11. Vishing: In this scam, victims' private data is taken from their phones. To trick victims into disclosing personal information and granting access to personal accounts, cybercriminals employ sophisticated social engineering techniques. Vishing connives to convince victims that they are being courteous by returning the call, much like phishing and smishing do. Many times, callers will pose as representatives of the government, the tax authority, the police, or the victim's bank.

12. Smishing: As the name implies, smishing is a type of fraud that takes advantage of text messages sent to mobile devices to convince its victims to contact a phony number, go to a phony website, or download dangerous software onto their computers.

13. Debit card fraud or credit card fraud: These crimes involve making unauthorized transactions with or withdrawals from another person's card in order to access their funds. Credit/debit card fraud is the use of a customer's account to make unauthorized purchases or cash withdrawals. When a criminal has access to a cardholder's debit/credit number or personal identification number (PIN), fraudulent conduct takes place. Hackers or dishonest employees may get your information.

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14. Salami assault: It is a method of money theft, meaning the hacker takes little sums of money. The harm is so slight that it goes overlooked. There are typically two sorts of salami attacks: penny shaving and salami slicing. In salami slicing, the attacker accesses a customer's data, including bank and credit card information, using an online database. The attacker gradually deducts little sums from each account. These amounts inevitably add up to significant amounts of money being stealthily withheld from the joint accounts.

COMMON CYBER CRIMES

Nowadays most people in the world are using mobile phones and the Internet in their daily activities, it is quite common for many internet users have face different types of threats. Let's see some of such cyber-crimes attacks.

CYBERCRIME TARGETING COMPUTERS AND MOBILES

Cybercrime targeting computers and mobiles is any criminal activity that either targets or uses a computer, a computer network, or a networked device. Cybercriminals may target computers and mobiles to steal personal or financial information, install malware, disrupt operations, or extort money from victims.

CYBERCRIME TARGETING COMPUTERS

A cybercrime is a criminal act in which a person targets a computer system or network of devices in an effort to commit fraud, steal data, or get illegal privileges. Technology is used to commit this kind of crime, which mostly happens online.

Cybercrime targeting computers encompasses a wide range of illegal activities that exploit vulnerabilities in computer systems and networks. These activities can have serious consequences for individuals, businesses, and even governments. These crimes can be motivated by financial gain, revenge, or simply for the challenge.

Although there are four main types of computer crimes, a single criminal transaction might involve numerous offenses.

- Theft of intellectual property or marketing data blackmail, or sabotage of operating systems and programmes are some crimes that target computers.
  All of these crimes include the use of a computer by the perpetrator to gather data or harm operating systems.
- The computer's processes, or its instrumentality, rather than the substance of actual files, are utilized to perpetrate the crime in the second sort of crime.
  This category of crimes includes telecommunications fraud, credit card fraud, and fraudulent use of automated teller machines.
- Another form of computer crime is one in which a computer is relevant to the crime but not necessary for it to happen but it is related to the crime act.

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Common types of cybercrime targeting computers include

1. Malware attacks
2. Phishing attacks
3. Denial-of-service(DOS) attacks
4. Man-in-the-middle(MITM)attacks

CYBERCRIME TARGETING MOBILES

Cybercrime targeting mobile devices has become increasingly prevalent as smartphones and tablets are integral to our daily lives. These crimes exploit vulnerabilities in mobile technology and can have serious consequences for individuals and organizations.

Here are some common types of cybercrime targeting mobiles:

- **Data Breach**
- **Unprotected Wi-Fi**
- **Internet spoofing**

- **Attacks by Phishers**
- **Spyware**
- **Unreliable cryptography**
- **Ineffective Session Handling**
- **Mobile phone theft**
- **Mobile viruses**
- **Phishing**
- **Mishing**
- **Smishing**
- **Vishing**
- **Spyware and stalker ware**
- **Bluetooth attacks**
- **Sim card swapping**
- **Location tracking**
- **App-based threats**

Let's go over some of the threats in detail.

- **Unprotected Wi-Fi:** Free Wi-Fi is very alluring to users, and if anyone connects to it, hackers may take your data. Never access private services, such as banking or financial transactions, using free wi-fi since there may be a potential of money theft.

- **Attacks by Phishers:** The majority of phishing assaults occur in emails and messages. When a user clicks on a dubious link there's a possibility that virus files will download, corrupting and hacking your devices and causing data loss. They may send a form for you to fill up with your private information in some circumstances.

- **Malicious websites and apps:** when you manually download an app from a website, there's a potential that it might achieve certain goals, including encrypting or stealing data.

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- **Weak Passwords:** There is a chance that someone else could access the data if the passwords for the mobile devices are weak. Privacy concerns and data leaks could emerge from this. Therefore, be sure to use secure passwords for any apps or mobile devices.

- **Threats to Iot:** mobile security as most of us are aware, wearable technology like· smartphones, watches, and other devices that connect to the internet quickly and effortlessly performs most tasks. If these gadgets are hacked, abuse of them could incur significant expenses.

- **Mishing:** Mishing is a combination of mobile phone and phishing. Mishing attacks are attempted using mobile phone technology.
  A typical mishing attacker uses call termed as vishing or message(SMS) known as smishing. Attacker will pretend to be an employee from your bank or another organization and will claim a need for your personal details.

- **Smishing:** Smishing is a type of phishing attack that uses text messages (SMS) to trick people into revealing sensitive information or clicking on malicious links. Smishing attacks are often disguised as messages from legitimate companies, such as banks, credit card companies, or shipping companies.

- **Vishing:** Vishing, or voice phishing, is a type of phishing attack that uses voice calls to trick people into revealing sensitive information credit card numbers or other related data used in ID theft schemes from individuals.

- **SIM swapping:** SIM swapping is a type of fraud where a cybercriminal takes control of a victim's SIM card number. This can be done by calling the victim's mobile carrier and pretending to be the victim. Once the cybercriminal has control of the victim's SIM card number, they can use it to access the victim's online accounts and steal their personal information.

CYBERCRIME AGAINST WOMEN AND CHILDREN

- Cyber-blackmail, threats, cyberpornography, publishing and disseminating obscene sexual content, stalking, bullying, defamation, morphing, and creating false profiles are frequent cybercrimes against women.
- According to data from the National Crime Record Bureau as of 2019, the overall number of cybercrime occurrences increased by 18.4%, but the proportion of cases involving women increased by 28%.
- 10 730 events, or 20.2% of the 52 974 incidents recorded in 2021, were crimes against women, according to the data.
- Karnataka, Maharashtra, and Uttar Pradesh all had smaller shares of incidents in 2021 (2,243, 1,697, and 958) respectively.

Women are more likely to be victims of the following crimes:

1. Sextortion: was the most prevalent cybercrime committed against women throughout the pandemic. The criminals started extorting money or sexual favors from their victims by exploiting their private photos or altered images as a form of blackmail. The offenders threatened women and demanded that they engage in sexual encounters or write letters to them as a way of venting their annoyance about the outbreak. They also felt emboldened to use their manipulated photographs to scare victims in order to extract money from them because they had little money.

2. Phishing: To make money during the lockdown, criminals send phony emails that contain links to specific websites in an effort to trick the victim into providing personal information like contact information and passwords or in order to infect the victim's device with harmful viruses as soon as the link is clicked. These emails and texts seem to be real. The attackers use the victim's bank account and other personal information to conduct shady transactions from the victim's account to their own.

3. Pornography: The perpetrators engaged in online sexual assaults against women during the pandemic, changing the victim's photograph and putting it in pornographic content.

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4. Cyberstalking: It involved, among other things, contacting the victim via social media sites or phone positions despite her obvious lack of interest, posting messages on the victim's page (often in a threatening manner), and persistently bothering the victim with emails and phone calls.

5. Internet hacking: People began reading the news online during the outbreak. False news and information are more prevalent than ever before. The women became the targets of cyber hacking after clicking fraudulent URLs. Their phones were infected with malware that downloaded all of their personal data, activated the microphone and camera, and captured their private images and videos. Then, crooks exploit these images and bits of information to commit extortion and other crimes.

6. Cybersex Trafficking: This type of trafficking differs from physical sex trafficking in that there is no physical contact between the victim and the offender. When a dealer broadcasts, records, or takes photos of the victim participating in sexual or intimate actions from a central location, they are engaging in cybersex trafficking. The dealer then sells the footage online to customers and sexual predators. Women have been coerced, deceived, and subjected to blackmail into engaging in cybersex trafficking, which is the sexual abuse of women.

FINANCIAL FRAUDS

- Online fraud and payments have received a lot of attention lately. Being cashless is easier for consumers to manage on the one hand, but 70% of these users have fallen victim to online fraud.
- In general, "financial fraud" refers to a circumstance in which someone loses money dishonestly or maliciously.
- After COVID-19, online payments have significantly increased globally. Due to the payment's inclusion of UPI and Internet banking, the number of cyber money frauds has increased.
- The RBI recognized cyber financial fraud on July 1, 2016, and a circular titled "Master Directions on Fraud- Classification and Reporting by Commercial Banks and Select FIs" was released.

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Financial fraud is classified by the RBI as

1. Misappropriation and criminal breach of trust
2. Cheating and forgery
3. Unapproved credit facilities extended for payment of rewards or illegal gratification
4. Fraudulent encashment through forged documents
5. Fraudulent transactions involving foreign exchange
6. Manipulation of books of account or through fictitious accounts
7. Property conversion or any other type of fraud.

Rule 2.2 of the Circular states that these acts must be interpreted in accordance with the Indian Penal Code, 1860, which classifies them as crimes. The government has developed a system to report financial cybercrime under the I4C scheme and given specific rules in this regard.

What does I4C stand for?

The Government of India launched the 14C scheme, also known as the Indian Cyber-crime Coordination Centre, in 2019 to combat the escalating cyber financial fraud. The Ministry of Home Affairs provided oversight for the introduction of this program.

The following are among the centers established for the identification and investigation of cyber crimes.

Matters related with:

1. National Cybercrime Threat Analytics Unit.
2. Cybercrime Ecosystem Management Unit.
3. National Cybercrime Reporting Portal.
4. Platform for Joint Cybercrime Investigation Team.

Coordination with implementing agencies for:

1. National Cybercrime Forensic Laboratory Ecosystem.
2. National Cybercrime Training Centre.
3. National Cybercrime Research and Innovation Centre.

Some points for prevention of financial fraud

1. **Be aware of the different types of financial fraud**
2. **Protect your personal information**
3. **Use strong passwords and keep them safe**
4. **Be careful about clicking on links in emails or text messages**
5. **Check your credit card reports regularly**
6. **Report any suspicious activity immediately**

SOCIAL ENGINEERING ATTACKS

The term "social engineering" is used to describe a wide range of harm behaviours carried out through interactions with other people. Users are duped into revealing critical information or committing security blunders via psychological manipulation.

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Social engineering is the practice of persuading others to reveal sensitive information. The kinds of information that these criminals are looking for can vary, but when a person is targeted, the criminals typically try to trick you into giving them your bank or password information, or they try to gain access to your computer to covertly install malicious software that will give them access to your bank and password information as well as give them control over your computer.

Social engineering tactics for attack

Attacks using social engineering can be carried out anywhere there is a chance of human interactions. The five most typical types of digital social engineering attacks are listed below.

1. Baiting: As the term suggests, baiting attacks use a false promise to spark a victim's curiosity or sense of greed. In order to steal their personal information or infect their systems with malware, they trick users into falling for a trap.

2. Scareware: believing Scareware bombards computer victims with bogus threats and misleading alarms. Users are tricked into either serving only infected with malware, which leads them to install software that scareware to profit the perpetrator, or is malware in and of itself. Other names include fraud ware, deception software, and rogue scanner software.
A typical scareware example is the legitimate-appearing popup of your browser while advertisements show up while you're browsing the web and say things like, "Your computer may be infected with harmful spyware programs." Either it offers to install the malicious tool for you it sends you to a malicious website where your machine is infected.

3. Pretexting: Pretexting here, an attacker gains knowledge by telling a string of cleverly constructed lies. The con is frequently started by a perpetrator who poses as someone who needs the victim's private information to complete a crucial task.
The attacker typically begins by gaining the victim's trust by posing as a coworker, police officer, bank or tax official, or any person with the authority to know something. Through queries that are allegedly necessary to verify the victim's identification, the pretext collects crucial personal information.

4. Phishing: Phishing scams, one of the most common forms of social engineering attack, are email and text message campaigns designed to make victims feel rushed, curious, or afraid. Then it prompts people to divulge private information, click on links to nefarious websites, or open attachments that are infected with malware.

Spear phishing is a method of sending phishing messages to a particular organization to gain organizational information for more targeted social engineering.
Whaling targets executives from the top management in the organization, usually from private companies (or high-profile people like celebrities).

Voice phishing (vishing) phone calls may be automated message systems recording all your inputs. Sometimes, a live person might speak with you to increase trust and urgency.

SMS phishing (smishing) texts or mobile app messages might include a web link or a prompt to follow up via a fraudulent email or phone number.

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Angler phishing takes place on social media, where the attacker mimics the customer service team of a trusted company. They interrupt your communication with a brand and turn the conversations into private messages, where they escalate the attack.

Search engine phishing attempts to place links to fake websites at the top of any search results. The advertisements will be paid or use valid optimization methods to manipulate search rankings. The links are given in email, text, social media messages, and online advertisements.

Watering hole attacks infect popular web pages with malware to affect multiple users at the same time. Carefully planning on the part of the attacker is required to find vulnerabilities of the specific sites.

Social Engineering Attack Prevention

The following advice can also help you become more vigilant about social engineering scams.

- **Don't open any attachments or emails from unknown sources:** You are not required to respond to emails from senders you don't know. Cross-check and confirm the news from other sources, such as over the phone or directly from a service provider's website, even if you do know them and have doubts about their message.
- **Use multifactor authentication:** User credentials are one of the most valuable pieces of information that attackers look for. In the event that the system is compromised, using multifactor authentication helps to ensure the security of your account.
- **Be skeptical of alluring offers:** If an offer sounds too good to be true, consider it carefully before taking it. You can immediately tell if you're dealing with a genuine offer or a snare by Googling the subject.
- **Update your antivirus and antimalware programs:** Make sure automatic updates are on, or establish a daily routine of downloading the most recent signatures. Regularly verify that the updates have been installed and run a system scan to look for any potential infections.
- **Use a strong password.** Each of your passwords must be unique and complex. You are using several types of characters, including uppercase, numbers, and symbols.

Secure Network Usage Habits

- Never let strangers connect to the main Wi-Fi At home or the workplace, access to guest Wi-Fi connections should be provided. It allows secure and secure access to your primary encrypted, password-protected connection.
- Always use a Virtual Private Network (VPN). VPNs are services that provide you with a private, encrypted "tunnel" over the Internet connection.

MALWARE AND RANSOMWARE ATTACKS

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Cyberattacks can take many different shapes. Attacks using malware, phishing, and ransomware are on the rise and can target both small businesses and huge corporations.

Malware is any software that is used to access IT systems without authorization in order to steal data, interfere with system functions, or harm IT networks in any other manner.

What is the intent of malware?

- **Intelligence and intrusion**
  Exfiltrates data such as emails, plans, and especially sensitive information like passwords.
- **Disruption and extortion**
  Locks up networks and pcs, making them unusable. If it holds your computer hostage for financial gain, it's called ransomware.
- **Destruction or vandalism**
  Destroys computer systems to damage your network infrastructure.
- **Steal computer resources**

Uses your computing power to run botnets, crypto mining programs (cryptojacking), or send spam emails.

- Monetary gain
  Sells your organization's intellectual property on the dark web.

Types of Malware

1. **Virus**
2. **Worms**
3. **Trojan horse**
4. **Spyware**
5. **Botnets**
6. **Keystroke loggers**
7. **Fileless malware**

Virus

Viruses are a subgroup of malware. A virus is malicious software attached to a document or file that supports macros to execute its code and spread from host to host. Once downloaded, the virus will lie dormant until the file is opened and in use. Viruses are designed to disrupt a system's ability to operate. As a result, viruses can cause significant operational issues and data loss.

Worms

A worm is a type of malicious software that rapidly replicates and spreads to any device within the network. Unlike viruses, worms do not need host programs to disseminate. A worm infects a device through a downloaded file or a network connection before it multiplies and disperses at an exponential rate. Like viruses, worms can severely disrupt the operations of a device and cause data loss.

Trojan horse

Trojan horse is a program in which malicious or harmful code is inside apparently harmless Programming or data in such a way that it can get control and cause harm.

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Spyware

Spyware is malicious software that runs secretly on a computer and reports back to a remote user. Rather than simply disrupting a device's operations, spyware targets sensitive information and can grant remote access to predators. Spyware is often used to steal financial or personal information. A specific type of spyware is a keylogger, which records your keystrokes to reveal passwords and personal information.

Keystrokes logger

Keystroke logging, often called keylogging, is the practice of noting the keys struck on a keyboard, typically in a covert manner so that the person using the keyboard is unaware that such actions are being monitored.

- Keystrokes logger is a quicker and easier way of capturing the passwords and monitoring the victims.

Botnet: Botnet is a term used for the collection of software robots, or Bots, that run autonomously and automatically, Bot is an automated computer program that can gain

control of your computer by infecting it with a virus or other malicious code that gives access. Your computer system may be a part of a botnet even though it appears to be operating normally. Botnets are often used to conduct a range of activities, from distributing spam and viruses to conducting DOS attacks

Fileless malware

Fileless malware is a type of memory-resident malware. As the term suggests, it is malware that operates from a victim's computer's memory, not from files on the hard drive. Because there are no files to scan, it is harder to detect than traditional malware. It also makes forensics more difficult because the malware disappears when the victim's computer is rebooted. In late 2017, the Cisco Talos threat intelligence team posted an example of fileless malware that they called dnsmessenger.

RANSOMWARE

Is a subset of malware that is characterized by attackers encrypting data or holding systems hostage until they receive payment or a ransom. Phishing is an internet scam that uses dishonest or misleading methods to get people to divulge their personal information. CISA provides a range of services and tools that people and businesses can use to defend themselves against different kinds of cyberattacks.

Types of ransomwares.

1. Scareware: It is a sort of ransomware that frequently fools users by pretending to detect malware on the victim's computer and showing a bogus alert message. These assaults frequently pose as antivirus software that demands payment to remove fake infections.
2. Screen lockers: These applications are made to keep the victim from accessing any files or data on their computer by locking them out. Usually, a notification requesting payment to unlock it is presented.
3. Encrypting Ransomware: Ransomware that encrypts files also goes by the name "crypto-ransomware." which locks the victim's files and demands money in exchange for a decryption key.
4. DDOS extortion: DDoS extortion is the threat of a DDoS assault being launched on the network or website of the victim unless a ransom is paid.
5. Mobile ransomware: As its name implies, this type of malware targets mobile devices like smartphones and tablets and demands money in exchange for the decryption of data or the ability to unlock the device.
6. Doxware: Though less frequent, this sophisticated ransomware variety threatens to disclose private, explicit, or sensitive data from the victim's computer unless a ransom is paid.
7. Ransomware-as-a-service (raas): Cybercriminals provide ransomware programs as a service (Raas)) to other hackers or cyberattacks who then use those programs to target victims.

Some of the ransomware variants

- Cryptolocker, wannacry, Ryuk, Bad Rabbit, Locky, NotPetya, REvil.

How malware and ransomware infects a system or device

- Phishing emails and other social engineering attacks.

- Exploiting Operating system and software vulnerabilities.
- Drive-by downloads.
- Removable media.

Steps to take people and organizations to protect themselves from malware and ransomware attacks.

- Educate employees about cyber security.
- Use strong passwords and enable two-factor authentication.
- Keep operating system and software up to date.
- Use security software.
- Back up data regularly.
- Avoid downloading software from unknown source.

ZERO DAY AND ZERO CLICK ATTACKS

A zero-day exploit is a kind of cyber security assault that happens the same day the manufacturer discovers a software, hardware, or firmware fault. The assault is known as a zero day exploit or zero-day attack since it has been zero days since the security hole was most recently exploited. Because the developer has not yet had an opportunity to remedy the defect, this type of cyberattack is regarded as risky. Large l, government agencies, firmware, hardware, the Internet of Things, users with access to important corporate data, etc. are typical targets of zero-day exploits.

In contrast to typical assaults, zer0-click attacks don't demand any action from the users they target, such as clicking a link, enabling macros, or running an executable. They are dangerous because they are highly developed, frequently employed in cyberespionage operations, and frequently leave very few tracks.

Once a device has been hijacked, an attacker has two options: either they will install surveillance software or, in a much more devastating manoeuvre, they will encrypt the files and demand a ransom. Users have limited options for self-defence because, in most zero-click attacks, victims are unable to determine when and how they were infected.

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Zero-day and zero-click attacks are often used by sophisticated attackers, such as nation-state actors and organized crime groups. These attackers have the resources and expertise to develop and exploit sophisticated vulnerabilities.

Measures to protect from zero-day and zero-click attack

- You should remove any programs that you don't even use from all of your devices, especially your messaging devices.
- You must also refrain from "jailbreaking" or "actively searching" your phone. It disables a number of security mechanisms built into IOS and Android.
- Frequently keep all of your devices in storage. You'll be happy to know that if your devices have ever been corrupted, it is possible to restore them to their original state.
- On all of your devices, make sure the operating system, firmware, and apps are updated. Apply security updates as soon as they are made available.

- Make secure passwords for each login you use.
- Pop-ups must be turned off in online browsers. Even if you still manage to appear, avoid clicking on them. Malicious entities frequently utilise pop-up windows to spread adware.
- If you're not confident you can manage them, only purchase apps from authorized retailers. Regular users may unquestionably feel safe utilising the method they have been using to review programs.
- Viewing your accounts requires multiple-factor authentication.
- Use a barrier. Every major OS release includes inbound firewalls, whereas every commercial router off the shelf includes NAT firewalls. Check to see if these are on. Every time you click on a broken link, they might be the most powerful force in the universe.
- Use security software: Security software, such as antivirus and anti-malware software, can help to protect against zero-day and zero-click attacks, but it is important to note that security software cannot detect all threats.
- Keep software up to date: Software updates often include security patches that can help to protect against zero-day and zero-click attacks. It is important to install software updates as soon as they are available.
- Be careful about what attachments you open and what links you click on: Zero-day and zero-click attacks can be carried out through phishing emails, malicious attachments, and drive-by downloads. It is important to be careful about what attachments you open and what links you click on, especially in emails from unknown senders.
- Use a firewall and intrusion detection system: A firewall and intrusion detection system can help to protect against zero-day and zero-click attacks by blocking malicious traffic.
- Educate employees about cyber security: Employees should be educated about the dangers of zero-day and zero-click attacks, and how to identify and avoid them.

CYBERCRIMINALS MODUS-OPERANDI

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Modus operandi, in general, refers to a criminal's technique for successfully committing a crime. Every Modus Operandi must have at least these three components

- Ensure the crime's success
- Preserve privacy
- Make effective emigration easier

- Their modus operandi (a particular way or method of doing something) often includes espionage (the practice of spying) and sabotage (destroy ,damage)
- They typically use sophisticated tools and techniques to gain access to their targets systems.

Cybercriminals employ a wide range of methods to carry out their illegal activities. These methods can vary in complexity and sophistication, and they often target individuals, organizations, or even governments.

1.1 Sending irritable messages

People are frequently taken advantage of by others who send obnoxious and harassing text messages, or multimedia information, such as MMS.

1.1.1 Text Message

- Unaware and innocent people frequently fall into the traps of cybercriminals when they get lottery SMS, prize money emails, phony employment promises, or fake mail offering admission to reputable universities.
- Bulk text messages that are annoying, insulting, misleading, and defamatory are frequently sent via mobile devices. The true source could not be fixed as a result.

1.1.2 Multimedia Messaging

- Mobile phones are frequently used in small groups to disseminate multimedia messages that malign someone's reputation.
- As evidenced by the Delhi MMS Scandal, pornography, offensive texts, and cyberbullying are becoming increasingly prevalent and well-liked.
- In secret and in isolated locations, vulgar videos are frequently recorded for later use against the victim.

1.2 Making Offensive Phone Calls

By calling them and disturbing them, offenders can also harass other people. Criminals frequently employ anonymous calls as a powerful tool for making extortion or threatening calls. The use of this form of contact by stalkers to harass women is common.

1.2.1 Landline/Mobile Calls

- The lack of Caller ids on landlines makes it difficult to quickly analyse an incoming call, giving cyber stalkers, cyberbullies, etc. An unfair edge.
- Unknown callers from fictitious numbers or from someone else's phone via line tapping pose a severe threat as well.

1.2.2 Web-based calls

- By using different websites to impersonate the mobile number, calls can be placed.
- Any fraudulent or bothersome calls are made in this manner in order to conceal the caller's true location. These calls are frequently made for terrorist activities, illegal drug trafficking, ransom demands, and blackmail.

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1.2.3 Overseas Calls

- Without the assistance of international agencies, cybercriminals who engage in forgery from abroad are difficult to track down.
- ISD calls are notorious for using obscenity in pleasure calls that are actually recorded messages designed to trick the caller into falling into a financial trap by requesting sensitive financial information

1.3 Data Theft

- **Data theft** refers to the changing of data's shape by the introduction, suppression, or corruption of the original data through dishonest methods in order to achieve an unfair advantage. These can be done in a variety of ways, including:
- **Data idling** is the act of maliciously altering data while it is being processed by the computer or just before.

- **Data Leakage:** This is the illicit copying of master file data from a computer for the purpose of demanding ransom, extortion, or any other fraudulent activity.
- **Data snooping** is the practice of remotely accessing files or digital data using a valid password or by breaking the password. Then, for a profit, this data is sold to others.

1.4 Identity Theft:

- This entails fraudulently using an individual's electronic signature, password, or other distinctive identifying characteristics to steal their identity.
- It consists of fraudulent transactions, online share trading schemes, e-banking offenses, credit card fraud, etc.

1.5 Theft of Intellectual Property

- It entails copying someone else's authentic, original work without getting their permission or giving them any credit.
- **Patent and Copyright Infringement:** This is a classic example of intellectual property theft in which someone makes copied work or uses another person's process for their own financial gain.
- Software piracy is when someone copies the original software with the intention of reselling it for a profit.
- **Reseller Piracy:** Original Hardware is offered alongside illegally obtained content.

1.6 WEB Page Hacking

Unauthorized access to any electronic communication medium is what is known as hacking. This technique modifies the original web page of a website by changing its appearance and file content, embarrassing any reputable company and maybe resulting in service denial and significant financial loss. The following methods can be used to accomplish these.

1.6.1 Cyberbullying: It relates to cyberbullying, which is when someone bullies another person over a digital medium on the basis of information that is important or valuable to them, or simply for the enjoyment of bugging the victim.

1.6.2 Cyberstalking: Stalking is the practice of following someone using the Internet or other technology. This phrase is frequently used interchangeably with online abuse or harassment.

1.6.3 Web jacking: Along with social engineering, this is another phishing tactic that can be deployed. Attackers establish a false website, and when a victim clicks on the link, a message stating that the website has moved and they must click another link displays. The victim will be routed to a false page if he clicks on the link that appears to be legitimate.

1.7 Email-Based Attacks

These older forms of communication are being supplanted by e-mails in today's technological age since they are quicker, more dependable, conveniently available from anywhere, and more practical. However, despite all its advantages, there are some disadvantages as well. People with evil intent utilise these powerful communication tools to harass others, crash their systems, or cause them to lose important data.

E-mail bombing: Cyber activists can potentially bring down the entire system by using specially crafted software to send emails to a specific recipient's email address.

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Spam: Spam is the indiscriminate, spontaneous bulk message sending of spam that takes advantage of electronic communications platforms. It may result in the system not functioning properly or perhaps crashing, which would result in the loss of data.

1.8 Malware Attacks

Malware is a computer program created to infiltrate or harm a computer system without the owner's knowledge. It contains:

- **VIRUS:** "Vital Information and Resources Under Siege" is what it stands for. It operates on a host file and either copies itself or renders the system inoperable.
- **RAT:** "Remotely Access Tool" is what it stands for. Without even the victim's knowledge, this program can be used to remotely operate the system.
- **Worms:** Worms are programs that continue to replicate themselves until all of the system memory has been used. However, it does not require a host file like VIRUS does.
- **Spyware:** Spyware is the term for the program used to remotely monitor the victim's computer and his actions.
- **Backdoors:** Backdoors are specialized software or programming codes that allow access to get beyond a computer's security measures.
- **Key loggers:** Key loggers are programs that record the keystrokes the victim makes on the keyboard.

Here are some tips to protect yourself from cybercriminals

- Be careful about what information you share online. Only share personal information with trusted sources.
- Be aware of unsolicited emails and SMS messages. If you receive a message from someone you don't know, or from a company you don't do business with, don't click on any links or open any attachments.
- Keep your software up to date. Software updates often include security patches that can help to protect you from known vulnerabilities.
- Use a strong password manager to create and manage unique, strong passwords for all of your online accounts.
- Enable multi-factor authentication (MFA) whenever possible. MFA adds an extra layer of security to your accounts by requiring you to enter a code from your phone in addition to your password.

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REPORTING OF CYBER CRIMES

As people spend more time online, crimes like credit card fraud, identity theft., and harassment that were formerly only committed in person are now also committed online. You can contribute to making the Internet a safer and more secure place for everyone by reporting crimes to the proper authorities.

How can I file a complaint about cybercrime?

- In order to handle reports and investigations into cybercrimes, the crime investigation team has established numerous cybercrime cells in various Indian cities. Presently, a dedicated cybercrime cell exists in the majority of Indian cities. You are always welcome to file a complaint with the department of crime investigation or the cyber

police. The first and most important step in enforcing penalties for cybercrime is to file complaints.

- A written complaint must be submitted to the local cybercrime unit. You must include your name, contact information, and mailing address in the written complaint. When registering a cybercrime complaint, you must address your written letter to the city's head of the cybercrime cell.
- The IT Act states that a cyber-crime is subject to global jurisdiction, so any cyber cell in India can receive a complaint about a cyber-crime, regardless of where the crime was first committed or where the victim is currently living or staying.

How can I file a police report for a cybererime?

- If none of India's cyber cells are accessible to you, you can submit a First Information Report (FIR) at the neighbourhood police station. If your complaint is rejected there, you might go to the Commissioner or the Judicial Magistrate of the city.
- The IPC covers some cybercrime offences. To report them, file a FIR with the local police station that is most convenient for you. No matter what jurisdiction the crime was committed in, every police officer is required by Section 154 of the Criminal Procedure Code to record the information or complaint of an offence.

How do I report a cybercrime online?

Cybercrime.gov. In/Accept.aspx is a government of India initiative that accepts complaints about online child pornography (CP), child sexual abuse material (CSAM), or sexually explicit content like rape/gang rape (CP/RGR), as well as other cybercrimes like social media crimes, online financial frauds, ransomware, hacking, cryptocurrency crimes, and online cyber trafficking. The portal also offers a method for anonymously filing a complaint on the reporting of content that contains rape/gang rape (RGR) or child pornography (CP).

The steps listed below can be used to report a cybercrime online:

STEP 1: Go to https://Cyber-crime.gov.in/Accept.aspx
STEP 2: Click on Report other Cyber-crimes' on the menu.
STEP 3: Click on 'File a Complaint'.
STEP 4: Read the conditions and accept them.
STEP 5: Register your mobile number and fill in your name and State.
STEP 6: Fill in the relevant details about the offence

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REMEDIAL AND MITIGATION MEASURES

A remedial measure in cybersecurity is an action taken to correct a vulnerability or weakness in a system that could be exploited by a cyber attacker.

Risk mitigation in cybersecurity

By implementing security rules and processes, a cybersecurity threat reduction strategy lowers the overall risk or impact of a cybersecurity incident. Prevention, detection, and remediation are the three parts of risk mitigation. Your organization's cybersecurity risk mitigation measures must change as cybercriminals modify their techniques to keep one step ahead of your organization's security risk mitigation procedures.

8 methods for reducing cybersecurity risk.

As the likelihood of facing a cyber assault is all but certain, proactive cybersecurity risk mitigation is swiftly replacing reactive cybersecurity risk management as the sole choice for organizations.

The following are the top 8 methods for reducing cybersecurity incidents throughout your IT ecosystem

1. Perform a cybersecurity risk analysis.
2. Implement network access restrictions.
3. Install firewalls and antivirus programs.
4. Schedule for patch management.
5. Continually keep an eye on network traffic.
6. Create an incident response strategy.
7. Examine your company's physical security.
8. Reduce your attack area.

LEGAL PERSPECTIVE OF CYBERCRIME

- Legal Perspectives, First Internet regulation Cybercrime can be defined as illegal conduct in which a computer is either a tool, a target, or both.
- Traditional criminal acts including theft, fraud, forgery, defamation, and mischief are all covered by the Indian Penal Code and can be included in cybercrimes.
- The Information Technology Act of 2000 addresses a wide range of new-age crimes that have been created as a result of computer abuse.

The Need for Cyber Law in India

- The world is becoming more and more digitally sophisticated, and so are the crimes, in today's technologically advanced society.
- The Internet was initially created in an unregulated manner as a tool for study and information sharing. With the development of e-business, e-commerce, e-governance, and e-procurement, among other things, it became more transactional over time.
- Cyber laws address all legal issues relating to online criminal activity.
- The demand for cyber laws and their implementation has grown significantly along with the increase in internet users.

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The need for cyber law can manifest itself in the following ways:

- Because all share-related transactions are now made in demat form, everybody involved in them needs access to the internet and protection from cybercrime.
- The majority of Indian businesses store their official data electronically. A corporation may require the aid of this law to prevent the misuse of such data.
- Due to the quick development of technology, many government forms, including ITR returns and service tax returns, are now filled out electronically.
- Anyone can simply misuse the forms by hacking into government portal websites. You can only obtain redress against this kind of fraud under cyber law.

- Both debit and credit cards are used by consumers when they shop. However, some online scammers duplicate those credit cards and debit cards.
- Card cloning is a method where someone can quickly access your credit card information using the internet. Cyber law can assist you in tracking down such crooks with ease.
- E-contracts and digital signatures are the two most popular ways to do business. Anyone connected to such digital signatures and e-contracts has the ability to easily commit fraud by abusing them. You are protected from this kind of scam by cyberlaw.

IT ACT 2000 AND ITS AMENDMENTS

IT ACT 2000

- The Infomation Technology Act, 2000, sometimes referred to as an IT Act, is a law that the Indian Parliament proposed and released on October 17, 2000
- The United Nations Model Law on Electronic Commerce of 1996 (UNCITRAL Model), which was recommended by the General Assembly of the United Nations in a resolution dated January 30, 1997, serves as the foundation for this information technology act.
- It is India's most significant law addressing cybercrime and e-commerce.
- This law's primary goal is to facilitate legal and reliable electronic, digital, and online transactions as well as to lessen or eliminate cybercrimes.
- There are 90 sections and 13 chapters in the IT Act. The Indian Penal Code of 1860 has been revised in the last four parts, starting with "section 91" and ending with "section 94."

It has two schedules

First schedule

Regarding documents or transactions to which the Act shall not apply, the first schedule is concerned

- Bills of exchange, promissory notes, and other instruments of a negotiable character

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The second schedule

The electronic signature or electronic authentication method is covered in the second schedule.

- The types of electronic signatures that are recognized as legally valid
- The requirements for creating and using electronic signatures
- The procedures for authenticating electronic records

The IT Act of 2000 lists the following offenses and penalties:

The following are the offenses and penalties covered by the IT Act of 2000

- Altering the source documents on the computer.
- Instructions from a controller to a subscriber to increase capacity for information decryption.
- The electronic publication of pornographic content.
- Penalties for violating privacy and secrecy as well as malicious hacking.
- Punishment for publishing a digital signature certificate that is false in some respects.
- Misrepresentation-related fines and confiscation.

- Investigative authority and a safe system.
- Fines for confiscation do not conflict with other sanctions.
- Apply for a crime or violation that was committed outside of India.
- A publication intended to commit fraud.
- The controller's ability to issue commands.

Sections and Punishments under the Information Technology Act of 2000 are as follows:

- **Section 43:** This clause of the Information Technology Act of 2000 specifies that any act of stealing, damaging, or changing a computer system or network, or deleting data without the owner's consent and with malice is punishable by payment to the owner as compensation for damages.
- **Section 43 A:** According to this clause of the IT Act of 2000, any business entity dealing with sensitive information that neglects to put reasonable security procedures in place and results in the loss of another person shall also be accountable as a criminal defendant for compensation to the affected party.
- **Section 66:** Hacking into a computer system with malign intent, such as fraud, carries a sentence of three years in prison, a fine of Rs.5,000, or both.
- **Section 66 B, C, D:** Identity theft, fraud, and dishonest use of information are all punishable by three years in prison, a Rs.1,00,000 fine, or both.
- **Section 66 E:** Violations of privacy through transmitting images or private areas are prohibited and are subject to a three-year prison sentence, a Rs.2,00,000 fine, or both.
- **Section 66 F:** This section discusses how cyberterrorism that compromises India's unity, integrity, security, or sovereignty online is punishable by a life sentence.
- **Section 67:** According to this clause, posting pornographic material or transmitting explicit material in public is punishable by up to five years in prison, a fine of Rs.10,00.000, or both.

Amendments

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- In 2008, a significant adjustment was made. The introduction of Section 66A made sending "offensive messages" illegal.
- Additionally, Section 69 was added, giving authorities the ability to "intercept, monitor, or decrypt any information through any computer resource."
- It also included laws addressing voyeurism, child porn, cyberterrorism, and pornography. Without any debate, the amendment was approved by the Lok Sabha on December 22, 2008. The Rajya Sabha approved it the following day. On February 5, 2009, President Pratibha Patil signed it into law.

CYBERCRIME AND OFFENCES

Cybercrime is a criminal activity that involves the use of electronic communication networks to commit illegal acts, either for personal gain or to harm others.

Cybercrime can have a devastating impact on individuals, businesses, and governments. It can lead to financial losses, identity theft, damage to reputations, and even physical harm.

A person commits the criminal offense of illegal access to a computer system when he or she accesses the whole or any part of a computer system without right.

The offenses included in the I.T. Act 2000 are as follows

- Tampering with the computer source documents.
- Hacking with computer system.
- Publishing of information that is obscene in electronic form.
- Power of Controller to give directions.
- Directions of the Controller to a subscriber to extend facilities to decrypt information from a protected system.
- Penalty for misrepresentation.
- Penalty for breach of confidentiality and privacy.
- Penalty for publishing a Digital Signature Certificate false in certain particulars.
- Publication for fraudulent purposes.
- Act to apply for offense or contravention committed outside India Confiscation.
- Penalties or confiscation not to interfere with other punishments.
- Power to investigate offenses.

ORGANISATIONS DEALING WITH CYBERCRIME AND CYBER SECURITY IN INDIA

- **The Indian Cybercrime Coordination Centre (I4C**) was established by MHA, in New Delhi to provide a framework and eco-system for Law Enforcement Agencies (leas) for dealing with Cybercrime in a coordinated and comprehensive manner.
- **National Cyber Forensic Laboratory:** This laboratory provides early-stage cyber forensic assistance to investigating officers of all state and union territory police.
- **Cyber Surakshit Bharat:** This is an initiative taken by the Indian government for cyber security.
- **The Indian Computer Emergency Response Team (CERT-In):** This is an initiative taken by the Indian government for cyber security.
- **National Critical Information Infrastructure Protection Centre (NCIIPC):** This is an initiative taken by the Indian government for cyber security.
- **NCCC (National Cyber Coordination Centre):** NCCC operates under the Ministry of Home Affairs and focuses on real-time threat assessment and situational awareness. It plays a crucial role in monitoring and analyzing cyber threats to ensure a secure cyberspace environment.
- **NCRB (National Crime Records Bureau):** NCRB is responsible for collecting and analyzing crime data, including cybercrime statistics. It serves as a central repository for information related to various types of crimes, including cybercrimes.
- **Cybersecurity and Information Security Research Centre (C-ISRC):** C-ISRC is a research and development organization established by the Ministry of Electronics and Information Technology (meity) to focus on cybersecurity and information security. It conducts research on various cybersecurity domains, develops cybersecurity solutions, and collaborates with industry partners to enhance cybersecurity capabilities in India.

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CASE STUDIES

Case 1: In the first cyber-defamation case in India, SMC Pneumatics (India) Pvt. Ltd. V. Jogesh Kwatra, an employee of the plaintiff company sent derogatory, defamatory, obscene, obscene, vulgar, filthy, and abusive emails to its employers as well as to various subsidiaries of the said company all over the world with the intention of defaming the company and its Managing Director. The Honorable Delhi High Court issued an ex-prate ad interim injunction, noting that the plaintiff had shown a prima facie case. Derogatory, defamatory, obscene, vulgar, degrading, and abusive emails were therefore prohibited by the Delhi High Court from being sent to the plaintiffs or to sister businesses around the world, including their managing directors and their sales and marketing departments. The Hon. Court further prohibited the defendant from discrediting, libelous, or abusively posting, sending, or causing to be published any material about the plaintiff in both the real world and online.

Case 2: In a different case, State of Tamil Nadu v. Suhas Katti2, it was alleged that someone posted an offensive, derogatory, and bothersome comment on a Yahoo messaging forum about a divorced woman. The offender constructed a phony email account in the victim's name and forwarded emails to the victim asking for information. The lady received bothersome phone calls after the message was posted because others thought she was begging. In response to a report the victim filed in February 2004, the police discovered the culprit was in Mumbai and promptly apprehended him. The Additional Chief Metropolitan Magistrate found the accused guilty of offenses under Sections 469, 509, and 67 of the IT Act, 2000, relying on the expert witnesses and other evidence presented before it, including the witnesses of the owners of the Cyber Cafe. The accused was found guilty and sentenced to two years of rigorous imprisonment (RI) under Section 469 of the IPC and a fine of Rs. 500, and to three years of rigorous imprisonment (RI) under Section 509 of the IPC.

Case 3: The Delhi High Court determined that publication is a broad term that encompasses all forms and mediums, including the internet, in the case of Tata Sons v. Turtle International. The fact that an internet publication has a larger audience, more permanency, and is more accessible than other tangible (as opposed to intangible) forms of expression does not change the fact that it is a forum or medium.

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Case 4: Sony PlayStation Network Hack (2011):
Background: In April 2011, the Sony PlayStation Network experienced a massive security breach.
Incident: Hackers gained unauthorized access to the network, compromising personal information, including names, addresses, and credit card details of millions of users.
Impact: Sony faced severe backlash, financial losses, and a tarnished reputation. The incident highlighted the vulnerability of large-scale online platforms to cyberattacks.

Case 5: WannaCry Ransomware Attack (2017):
Background: In May 2017, the WannaCry ransomware spread globally, affecting hundreds of thousands of computers in over 150 countries.
Incident: The ransomware exploited a Windows vulnerability, encrypting users' files and demanding ransom payments in Bitcoin for decryption keys.

Impact: The attack affected critical systems, including healthcare, transportation, and businesses, emphasizing the importance of regular software updates and cybersecurity measures.

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UNIT – 3

SOCIAL MEDIA OVERVIEW AND SECURITY

INTRODUCTION TO SOCIAL NETWORKS

What are social networks?

- Social media is a channel of communication that uses the internet. Users can engage in conversations, share information, and produce content for the web via social media platforms.
- Social media can take many different forms, including blogs, microblogs, wikis, social networking sites, photo and video-sharing websites, instant messaging, podcasts, widgets, virtual reality, and more.
- To share ideas, interests, and activities, people and organizations are grouped through social networking.
- There are several web-based social network services, like facebook, Twitter, linkedin, google+, etc., that provide a simple and dynamic interface for connecting with individuals both domestically and abroad.
- Additionally, there are a number of mobile-based social networking services available as apps, including Whatsapp, Hike, Line, and others.
- For users with a wide range of interests, social media includes numerous distinct sorts of networks. Social media marketing and advertising are cost-efficient and successful, enabling you to reach a larger audience. Understanding the most common social media platforms will enable you to develop more precise, more targeted marketing strategies.

TYPES OF SOCIAL MEDIA

Here are some of the most popular types of social media

1. **Social media**

- Social networking platforms facilitate interpersonal connections and provide a plenty of opportunities for various brands to draw customers. Users can curate information submit photographs and videos, form groups based on their interests, express their ideas, and take part in group conversations. ·

  You may use the information you get about what people are saying about your company on social media to build a strong presence and raise brand recognition.

  Social networking sites include for instance: **Instagram, Twitter, TikTok, LinkedIn.**

2. **Forums for discussion**

- Discussion boards empower users to discuss news, opinions, and answers to each other's questions. In order to get the best response, many of these social media platforms concentrate on asking questions. Your trust in the forum can rise if you respond to them truthfully and accurately.
- Discussion forums examples: **Digg, Reddit, Quora, Clubhouse.**

3. **Networks for sharing images:** People can exchange pictures and relevant content on these social media platforms. They provide a forum for generating ideas, encouraging

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innovation, enhancing the appeal of your products, and getting people to talk about your company. For instance, a clothing firm might offer a chance to win a gift card to anyone who uploads a photo of their stuff to the image-sharing website. A few instances of image-sharing websites are **Flickr Instagram and Photobucket**

4. **Networks for bookmarking**: Users can save various thoughts, articles, blogs, and other stuff on bookmarking networks for later use. Links to online resource lists are also frequently shared. These websites' objectives are to discuss trends and find fresh content based on common interests. By sharing your own material on these networks, you may attract visitors to your website, engage with them, and shape their opinions.
   Networks for bookmarking include **Feedly, Pocket, Flipboard, Stumble Upon, and Pinterest.**

5. **Networks for blogging and publishing**: You can share your opinions on a variety of topics on these social media platforms, including your profession, current events, hobbies, and more. Many advantages of running your own blog are available to you even if your website doesn't host it.
   To increase traffic to your account, think about publishing blog posts on inventive methods to use your product or posting news about your sector. Examples of publishing networks and blogs are **Medium, Facebook, WordPress, and Tumblr.**

6. **Networks for user reviews**: These websites showcase client feedback, providing readers with a thorough understanding of the kinds of services and goods provided as well as the general satisfaction level. Reviews from customers can help you spot any issues that many people encounter frequently and provide you the opportunity to enhance their experiences over time.
   For instance**: TripAdvisor, OpenTable, Yelp, My Business on Google**

7. **Networks based on interests**: Use these specialised networks to let others know about your interests and activities. You can network based on shared interests by using websites and applications linked to your sector. People might be able to discuss their favorite books and obtain suggestions for what to read next on an interest-based website, for instance.
   Interest-based networking examples include **Peanut, Strava, and Goodreads.**

8. **Shared economy networks**: By encouraging people to exchange goods and services, these websites enable access to resources. For instance, a pet-specific sharing economy website may assist a user in finding someone willing to look after their dog while they are away.
   Examples of networks in the sharing economy: **Couchsurfing, Lending Club, Eatwith**

9. **Social commerce platforms:** These networks assist users in identifying trends, sharing amazing finds, making purchases, and following their preferred businesses. They concentrate on online shopping, and the social component makes it interesting and fun. These social shopping platforms are excellent for increasing brand recognition and expanding your customer base.
   Social purchasing networks include, for example: **Poshmark, Instagram, Etsy, and Facebook.**

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10. **Websites that host videos:** Independent filmmakers, journalists, and other artists can stream videos rapidly and easily thanks to platforms for hosting videos. Brands can shoot their own video content, encourage influencers to use and discuss their products, or utilize paid advertisements to attract new customers.
    Several websites that host videos include **YouTube, Tinder, Snapchat, YouTube, and Instagram.**

SOCIAL MEDIA PLATFORMS

- It's useful to be aware of the top social media platforms at the moment, whether you're an experienced social media marketer, a marketer trying to get started with social media marketing, or a business owner looking to use social media.
- By doing this, you'll be able to engage with the relevant individuals, increase the reach of your business on social media, and accomplish your social media objectives.
- Of course, it's not just a matter of how big the social media platforms are.
- Consider whether you and your company are a good fit for the social networking site. Does it fit the image of your brand? Does your target market use that social media platform? How many social media platforms can you simultaneously manage?

Here are some of the most widely used and active user's social media platforms:

1. **Facebook has 2.96 billion Monthly Active Users (MAUs)**
   - With about 3 billion monthly users, Facebook is the most popular social networking platform. This indicates that 37% or so of people on Earth use Facebook.
   - If you want to have a presence on social media, Facebook is a reasonably secure pick because more than 200 million businesses (mainly small businesses) utilize its capabilities and more than seven million marketers actively promote their company there.
   - Facebook is simple to use because practically all content formats, including text photos, video material, and Stories, function well there. However, Facebook always gives priority to content that promotes meaningful interactions and conversations among individuals, particularly those from family and friends.

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2. **YouTube, which has 2.2 billion MAUs.**
   Every day, users of the video-sharing website YouTube watch one billion hours of content. In addition to being the second-largest social media platform, YouTube parent company of Google is frequently referred to as the second-largest search engine. Therefore, if you utilize video to market your business, you must include YouTube in your marketing

3. **WhatsApp has 2 billion MAUs**
   - People use WhatsApp a messaging service, in more than 180 different countries-. Initially, WhatsApp was primarily used by users to text their relatives and friends. People gradually began contacting companies using WhatsApp.
   - Businesses may offer customer service and update clients on their purchases using WhatsApp's business platform.

- The WhatsApp Business app is available for small businesses, and the WhatsApp Business API is available for enterprise businesses.

4. **Instagram has 2 billion monthly users**
   - Instagram is a great venue for showcasing your goods or services because it's a visual social networking site. You can share a variety of media on the app, including images, videos, Stories, reels, live videos, and longer-form movies on IGTV.
   - You can set up an Instagram business profile as a brand, which offers detailed analytics for your profile and posts and the capacity to plan Instagram updates using third-party applications. Because consumers post content and tag businesses frequently, it's also a terrific location to gather user-generated content from your audience.

5. **WeChat has 1.26 billion MAUs**
   - Tencent, one of the largest technology corporations in China, introduced WeChat in 201l. WeChat began as a messaging program, similar to WhatsApp and Messenger, but has since transformed into a multipurpose platform.
   - Users can shop online, pay bills, buy groceries, send money, and make bookings, phone, and more in addition to chatting and calling.
   - In China and other areas of Asia, WeChat is the most widely used social media platform. WeChat is a wonderful alternative if you want to promote your company in China (where big social networks like Facebook and Twitter are prohibited).
   - You can run advertisements on WeChat Moments or the official WeChat account. Additionally, a lot of influencers with a WeChat following might assist your business in reaching millions of Chinese consumers.

6. **TikTok has 1 billion MAUs**
   - TikTok is a short-form video-sharing app (also known as Douyin in China). Despite only being introduced in 2017, it is one of the world's fastest-growing apps and recently surpassed Google as the most popular website.
   - With TikTok users can generate and share 15-60-second videos that can be enhanced with a wide variety of sound effects, music clips, and filters to make them more visually appealing. Videos relating to virtually all interests are available, from lip-syncing, dancing, and challenges to do-it-yourself tips and makeup instructions.
   - Ages 10-29 make up about 47.4% of TikTok users in the United States. So TikTok is a terrific platform for your business if your target market is young.

7. **Telegram has 550,000,000 MAUs**
   - With no restrictions on media sizes, Telegram is free chat software that functions on many different devices.
   - End-to-end encryption for all operations, including conversations, groups, and participant-shared media, is what makes Telegram stand apart from the competition.
   - Since WhatsApp announced revisions to its privacy policy that would allow it to share information with parent firm Meta, its emphasis on security has attracted more users.

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8. **Snapchat has 557 million MAUs.**
   - Snapchat focuses on sending snaps-short movies and photos to pals. It helped popularise the vertical video format, which later spread to other social media sites like Instagram and TikTok.
   - However, the popularity of Instagram Stories in particular appears to have hampered Snapchat's development and. more generally, advertisers' enthusiasm for using Snapchat for their business.
   - However, 69% of American teenagers claim to use Snapchat. So, you should think about using the app if your primary target demographic is adolescents.

SOCIAL MEDIA MONITORING

Social media is now used by the majority of e-commerce platforms to interact with both current and potential clients. Businesses may communicate with customers in real time thanks to these networks. Monitoring social media can help you understand how your business is being discussed online.

What is social media monitoring?

Monitoring social media is finding out what is being said about a company, person, or product on various internet and social media platforms.

- The act of gathering social media conversations and messages into a database of informative data is known as social media monitoring. Additionally, you have the option to respond to each message you read separately, addressing issues, grievances, and inquiries on each channel.
- The process of social media listening involves analyzing the insightful conversations you record. From there, you may expand your campaigns and alter your company in response to trends you identify in the data.

The advantages of social monitoring

The majority of social media monitoring solutions provide social listening capabilities. These solutions can build your brand and improve your communication strategy when they are utilized properly.

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The following are some advantages of social media motoring:

- **Increase brand reputation:** By staying informed about what your consumers are saying, you can quickly address their demands.
- **More engaged Customers:** Monitoring identifies possibilities for firms to engage in beneficial dialogues with customers. Over time, this results in stronger partnerships.
- **Better campaign insights:** On tools like Sprout Social, you can tag messages and use keywords to track how various audiences react to your brand's mixed messaging.
- You won't have to be concerned about missing out on changing trends in customer experience or sentiment because you'll always be in the loop. You can always be aware of what your audience is thinking and feeling by using social media monitoring.

Some of the social media monitoring tools are

- **Sprout:-** With Sprout Social, you have all the tools you need to both track and respond to crucial interactions. To begin with, the software monitors mentions, comments, and keywords across numerous social media platforms, including Instagram, Facebook, Twitter, and TikTok.

- **Hootsuite:-** Hootsuite is a social media management tool that monitors every major social network, both paid and unpaid. It offers a free version with basic features and premium plans for advanced functionalities. The platform's configurable search streams allow users to track keywords, hashtags, regions, and individuals across multiple social media sites. Known for its simplicity, Hootsuite provides an easy-to-use interface for monitoring trends and mentions at a glance. It is a comprehensive solution for managing and analyzing social media activities.

- **Mentionlytics:-** Mentionalytics is a social media analytics tool designed to track and analyze online mentions. It provides insights into brand mentions, keywords, and trends across various platforms. This tool helps businesses understand their online presence and audience engagement by monitoring conversations. Mentionalytics offers customizable dashboards and reporting features for data-driven decision-making. With its focus on real-time analysis, it empowers users to stay informed about their online reputation and social media impact.

- **Reputology:-** Reputology is a reputation management platform that enables businesses to monitor and manage online reviews. It consolidates reviews from various platforms, providing a centralized view for efficient analysis. The tool helps businesses track customer sentiment, respond to reviews, and enhance their online reputation. With features like real-time alerts, Reputology allows proactive engagement to maintain a positive brand image.

- **Tweepsmap:-** Tweepsmap is a social media analytics tool tailored for Twitter. It visualizes the geographical distribution of a Twitter account's followers, helping users understand their audience's location. With features like follower demographics and engagement analytics, Tweepsmap assists in strategic Twitter content planning.

- **Reddit Keyword Monitor Pro:-** Reddit Keyword Monitor Pro is a monitoring tool designed for Reddit, allowing users to track specific keywords or topics across the platform. It provides real-time alerts and notifications for mentions, enabling users to stay informed about relevant discussions. With advanced filtering options, it aids in efficiently monitoring and engaging with specific content on Reddit.

- **Talkwalker:-** Talkwalker is a powerful social media analytics tool for businesses, offering real-time monitoring and sentiment analysis across platforms. It provides customizable dashboards and competitive intelligence, empowering data-driven decisions for effective brand management and social media strategy.

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HASHTAG

Hashtags are keywords or phrases preceded by the "#" symbol used to categorize and organize social media content. They help users find content related to their interests and connect with others who share their passions. Hashtags are commonly used on platforms like Twitter, Instagram, TikTok, and Facebook.

Hashtags can be used in a variety of places on social media, including

- **Posts:** Hashtags can be included in the body of a post to help people find it.
- **Profiles:** Hashtags can be added to a user's profile to help people find content related to their interests.
- **Comments:** Hashtags can be added to comments to help people find them.
- **Direct messages:** Hashtags can be added to direct messages to help people find them.

Hashtags are important to

- **Increased visibility:** Hashtags make your content more discoverable by users who are searching for specific topics or keywords.
- **Community building:** Hashtags can help you connect with like-minded individuals and foster a sense of community around shared interests.
- **Trend identification**: Hashtags can reveal current trends and topics of conversation, allowing you to stay up-to-date and participate in relevant discussions.
- **Brand awareness:** Hashtags can be used to promote your brand, products, or services and reach a wider audience.
- **Campaign tracking:** Hashtags can be used to track the performance of social media campaigns and measure engagement levels.

VIRAL CONTENT

A post video or image that has gone "viral" on social media has gained enormous popularity and is being widely shared on a variety of social media platforms.

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What does it mean to go viral?

- Going viral can happens naturally when viewers are drawn to the material and automatically share it with their followers, or it can happen as the consequence of a carefully planned and carried out marketing campaign.
- There are many different types of viral material, such as videos, memes, photos, and text posts. It might be light-hearted and enjoyable or serious and newsworthy.
- A brand or person can be significantly impacted by going viral, which can increase visibility, reach, and engagement. The need to monitor and respond to a significant number of engagements, negative comments, and the potential for the information to be misunderstood or harmful can all be difficult.

How do you become a social media sensation?

Social media success takes a trifecta of serendipity, creativity, and strategy.

Here are some pointers to improve your likelihood of becoming viral.

1. Understand Your Audience
To create content that has the potential of going viral, it is essential to understand your target audience and the kind of content they are likely to engage with.
2. Produce distinctive and captivating content
Your material must be original, fascinating, and engaging if you want it to go viral. This can include being adventurous and creative, but it's crucial to make sure that your content is consistent with the principles and message of your business.
3. Make use of popular hashtags and themes.
Your content's visibility and reach can be increased by keeping up with hot subjects and using pertinent hashtags.
4. Work with influential people
Your content's visibility and reach can be increased by working with influencers who have a sizable following, which increases the likelihood that it will become viral.
5. Distribute on many platforms
Sharing your material on various social media sites will help it become more visible and reach more people.

Advantages of viral videos
For both people and companies, becoming viral has many advantages, including:

- **increased awareness**
  You or your brand may receive a lot of publicity and attention if it becomes viral. This could broaden your readership and help you reach a larger audience.
- **increase in credibility**
  A positive message or piece of content going viral can help you gain reputation and position yourself as an expert in your profession.
- **More participation**
  Increased involvement with your content, such as likes, comments, and shares, might result from going viral. This can improve your relationship with your audience and increase the size of y·our following.
- **increased brand awareness**
  Your target audience may be more likely to recognise and remember your brand if itis connected to a viral campaign or piece of content.
- **Increasing sales**
  Going viral can enhance sales and revenue for businesses, especially if the ~al material is relevant to a good or service you provide.

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SOCIAL MEDIA MARKETING

Utilizing social media networks to market goods, services, brands, or information to a specific audience is known as social media marketing.

- Due to the extensive use of social media platforms like Facebook, Instagram, Twitter, Linked, Pinterest, TikTok, and others, it has evolved into a crucial component of the total marketing mix for both organizations and individuals.

Social media marketing advantages

Without giving any thought to the results, many businesses leap into social media marketing or create a social media marketing plan. Without knowing the advantages, it doesn't make sense to spend time, effort, or money on social media.

You may assess the cost-benefit ratio and adjust your plan for the best results with a thorough understanding of the possible outcomes.

Here are just a few potential advantages:

- **Increased brand and business awareness**
- **Increased website traffic and**
- **Sales leads**
- **A rise in sales**
- **A chance to better understand customers**
- **Thanks to good customer engagement**

Frequently used social media platforms and tools:

- **Blogs:** A venue for casual conversation and debate on a particular subject or viewpoint.
- **Facebook:** As of the third quarter of 2015, Facebook had more than 1.55 billion monthly active members, making it the largest social network in the world. Users set up personal profiles, add friends, and communicate via messages and status updates. Facebook users can "like" the pages that brands create.
- **Twitter:** A social networking/microblogging site that enables communication between people and groups through the exchange of 140-character status updates.
- **YouTube and Vimeo:** Websites for hosting and viewing videos.
- **Flickr:** Flickr is a website and online community that hosts images and videos. On Facebook, Twitter, and other social networking websites, photos can be shared.

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- **Instagram:** A free picture and video-sharing program that enables users to edit their images with digital filters, frames, and effects before sharing them on a number of social networking platforms.
- **LinkedIn:** LinkedIn is a platform where groups of professionals with related fields of expertise can communicate and share information.

SOCIAL MEDIA PRIVACY

The term "social media privacy" describes the private and delicate information that can be obtain from your accounts about you. This data may be intentionally shared (in public profiles and postings, for example) or unintentionally shared (for example, websites sharing data with other businesses and social media marketing agencies).

- The ability to share content online has never been simpler thanks to social media platforms like Facebook, Instagram, and Twitter.
- But compared to other forms of online communication, publishing anything on social media is a little different.
- The information you publish on social media is more public than it is on email or instant messaging, which means that many more people will typically see it.

Advertisers and con artists can obtain sensitive information, such as the following, even if an account is private.

- **Updates on job, personal, and romantic happenings.**
- **Religious convictions.**
- **Information about the profile, such as name, contact details, and birthdate.**
- **Location information, such as your home city, check-in locations, prior residence cities, or precise address.**
- **Personal preferences, including website interactions and purchasing history.**
- **Content that is shared, such as status updates, pictures, and videos.**
- **Participation in social media, such as liking, sharing, or commenting on other posts.**
- **Information on employment, such as present or previous occupations.**
- **Personal identifiers like gender, race, and age.**

Think before sharing

- It's crucial to understand that social media platforms are fundamentally more open and sociable than conventional communication tools, even though they provide privacy options to assist you in restricting who can see the items you publish.
- Whether or not you are aware of it, anything you post online has the potential to influence how other people view you. You should always give serious thought to the information you disclose on social media.

Take a look at your privacy settings

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All social networking sites include privacy options that let you choose who you share information with.

- For instance, anytime you share something on Facebook, you have the option of sharing it privately with a select group of people, publicly with everyone on Facebook, or with all of your Facebook friends. Understanding how your privacy settings operate and how to modify them is crucial for this reason.

SOCIAL MEDIA CHALLENGES

Social networking is being used more frequently by business and governmental organizations to promote internal and external communication and teamwork. Increased communication has undeniable advantages, but social networks also come with a variety of drawbacks, including the following:

- **Consumption of bandwidth and storage:** A lot of users upload enormous data to social networks, including photos, videos, music, and movies in high quality. Your infrastructure could be severely harmed by downloading and keeping these files, which would make capacity planning nearly impossible.

- **Potential for legal trouble:** To harass a librarian they didn't like, students at Canterbury's University of Kent started a Facebook page called "For Those Who Hate the Little Fat Library Man." Employees could be held liable in any ensuing legal action if they utilize business, and IT resources for comparable reasons in the U.S.

- **Malware exposure:** Social networks are intended to be open platforms with little limitations on links or content. Security was typically not a top design priority. As a result, these networks might be used to spread spyware, worms, and viruses.

- **Decreased output of the workforce:** Personal social networking can have an impact on workplace efficiency. A trader for Goldman Sachs in the UK was using Facebook for four hours every day of work. When instructed to stop, he posted the warning email and added the following statement: "It's a measure of how warped I've become that, not only am I surprisingly proud of this, but losing my job worries me far less than losing Facebook."

- **Personal data disclosure:** Businesses frequently search MySpace, Classmates.com, Linkedin, and other social networking sites to learn more about possible employees and rivals, but postings should never be taken at face value.
- **Cyberbullying:** Social media provides a platform for communication, but it can also be a space for cyberbullying. Harmful comments, harassment, and online bullying can negatively impact individuals' mental health and well-being.

- **Fake News and Misinformation:** Social media can be a breeding ground for the spread of false information. Misleading content can go viral quickly, influencing public opinion and creating challenges for reliable information dissemination.

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- **Addiction and Mental Health:** Excessive use of social media has been linked to issues such as addiction and negative impacts on mental health. The constant comparison with others, cyberbullying, and the pressure to curate a perfect online image can contribute to stress and anxiety.

- **Polarization and Echo Chambers:** Social media algorithms may contribute to the formation of echo chambers, where individuals are exposed to content that aligns with their existing beliefs. This can lead to polarization and a lack of exposure to diverse perspectives.

OPPORTUNITIES AND PITFALLS IN ONLINE SOCIAL NETWORK

Opportunities

The internet has provided social -media the chance to flourish. Participating in social media forums also referred to as "Web 2.0," enables instantaneous communication and information sharing with large groups of individuals.

The opportunities in social media can be categorized into one of six main groups:

- **Social network:** Social networking sites are places where users may communicate and share with others who share their interests and history. Consider **Facebook.**
- **Bookmark:** Websites that let users save and organize connections to a wide range of online resources and websites are known as bookmarking sites. **StumbleUpon** is an example.
- **Publish news:** Sites that enable users to publish news links and other content to external articles are known as social news sites. Consider **Reddit.**
- **Media Sharing:** Websites that let users share different kinds of media, like images and videos, are known as media sharing sites. Consider YouTube.
- **Microblogging:** Microblogging refers to websites that let users post brief written entries that may contain links to websites offering goods and services as well as links to other social media platforms. Consider **Twitter.**
- **Blog Comments and Forums:** Websites that allow users to participate in conversations by posting and reacting to messages from the public are blogs and forums.

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Pitfalls in online social network

- **Too Many participants:** There are too many participants in the market for the several white-label social networks (white label meaning you can rebrand and establish your version of Facebook).
- **Users Exit as Marketers Move In:** Recall Friendster? Tribe or E-Circles from way back? There is nothing novel about the fact that when groups grow, marketers will enter and, in some circumstances, ruin the atmosphere, while the cool, hip influencers will go on to the next network.
- **Unreliable Member Information:** Users frequently (I've heard reports of up to one-third) provide false information in their profiles. Because of this, marketing initiatives won't be directed at the appropriate groups, and members will continue to be a moving target.
- **Lack of stats Makes Success Difficult to Measure:** Access to server stats isn't always available for many marketers who wish to launch a campaign on a social network. As a result, they frequently have to visually track user activity on the site or

count clicks to their website. A basic dashboard is offered in some of the more advanced platforms.

- **Stalkers and Other Unwanted Activity Ruin Lives:** MySpace continues to be an issue for child stalkers, and in some circumstances, it is simple to disguise oneself as someone else and easily fool others. As a result of the deceit, rejection, and humiliation from a peer's mother, one young teen committed suicide.
- **Privacy Issues:** The latest Beacon experiment, the introduction of newsfeeds, and social networks sharing too much information with third-party widget developers put users at risk and make them uncomfortable. As developers move in, tensions between users and developers have grown.

SECURITY ISSUES RELATED TO SOCIAL MEDIA

With social media platforms offering a quick method to interact with friends, family, and co-workers, exchange news and updates, and explore a wide range of hobbies, these platforms have become an essential part of our lives. But social media's broad use has also brought up several security issues that users need to be aware of.

The most frequent security threats on social media

- **Scams and phishing attacks:** Phishing scams are among the most practiced cyber security concerns involving social media. The objective of a phishing scam is to trick you or your staff into providing passwords, financial information, or other sensitive information.

- **False accounts:** It's not difficult for a fraudster to set up a social media account that impersonates your business. This is one of the key benefits of obtaining social network verification.

- **Attacks using malware and hacks:** If hackers access your social media accounts, they could seriously harm the reputation of your company. There is considerably more risk if they are successful in installing malware.

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- **Risky third-party applications:** It's a good idea to secure your personal social media accounts. Although there may still be security flaws in connected third-party apps that allow hackers to access your encrypted social media.

- **Identity theft:** online social media challenges and quizzes are a popular way to learn passwords or obtain personal information that is frequently used as a forgotten password hint.

- **Data security and privacy options:** It appears that most people are aware of the potential privacy dangers associated with utilizing social media. Overall confidence in social networks' capacity to safeguard data security and privacy has been eroding in recent years. TikTok has recently made headlines in particular because governments all over the world have restricted access to the platform on official devices due to data security concerns.

- **Mobile devices without security:** A mobile device without password protection is a gold mine for anyone with malicious intent. More than half of our internet time is spent on mobile devices. With just one swipe, social media apps make it simple to access your social network accounts.

To protect yourself from these security risks, it is important to be aware of them and take steps to mitigate them. Here are some tips for staying safe on social media:

- **Be mindful of what you share**
- **Be careful about the links you click**
- **Use strong and unique passwords**
- **Enable two-factor authentication**
- **Report if you find any suspicious activity**

FLAGGING AND REPORTING OF INAPPROPRIATE CONTENT

In the context of social media, "flagging" is the act of informing administrators or moderators of information that violates the platform's community standards, terms of service, or content restrictions. Users can "flag" content to bring it to the attention of the platform's moderation team when they come across posts, comments, profiles, or other content that they think is improper, hurtful, or in violation of the platform's guidelines.

Maintaining safe and courteous online communities requires flagging and reporting harmful content here are some broad guidelines you can adhere to flag and report problematic information on any online area, including social networking platforms, forums, websites, and other online spaces:

1. Recognise Community rules: Familiar yourself with any applicable community rules, terms of service, or content policies for the site. These policies often list the kinds of content that are unacceptable or forbidden on the platform.

2. Find Options for Reporting: The majority of sites provide you the option to report content right from the post or page where you find it. Attempt to find options such as "report," "Flag," or "Report Inappropriate Content." These are typically represented by an icon of a flag or some symbol.

3. Select the Correct Category: Platforms frequently require you to provide a justification for reporting information. Harassment, hate speech, lewd or sexually explicit material, violence, copyright violations, and spam are examples of typical categories. Choose the category that most accurately captures the problem you're reporting with the material.

4. Offer Details: You might be requested to offer particular information about the offensive content in addition to choosing a category. Be clear and detailed as you can when describing how the information is against the platform's rules.

5. Submit the Report: After you've completed the required fields, submit the report. You might be able to report numerous pieces of content simultaneously on some platforms.

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6. Review Confirmation: You can get a confirmation message after reporting the content. Your report has been received and is being examined by the platform's moderation staff, according to this confirmation.

7. Keep an eye on the situation: The moderation team may need some time to review and respond to the reported content, depending on the platform and the seriousness of the problem. If the content is damaging or inappropriate, be patient and don't engage further.

8. Follow-Up: If you think the problem is not being resolved or if you come across more offensive content, think about contacting the platform's support or moderation team again. You can check the status of your report on some platforms.

Keep in mind that content filtering can differ from platform to platform in terms of efficiency and speed. It's critical to report improper content for the safety and well-being of the larger online community as well as for your protection. You can help make the internet a safer and more civil place by reporting inappropriate information

LAWS REGARDING POSTING OF INAPPROPRIATE CONTENT

Social media laws in India: The Indian Constitution 1950:
Every person in India has freedom under the constitution, which the government cannot restrict. Article 19(1)(a) of the constitution provides freedom of speech and expression to every person but this right is not absolute, the government has imposed certain restrictions on them under Article 19(2) of the Indian constitution. The freedom of the press is implied throughout article 19(1)(a), while it is not specifically mentioned.

Information Technology Act 2000:

- Any offense done via social media to harm or cause injury to others is prohibited and punishable under Section 66(a) of the Information Technology Act of 2000.
- Section 69(a) states that the government has the authority to prohibit or stop the public from accessing material that is in conflict with government regulations. It also outlines the process for blocking public access to information that does not adhere to government regulations.

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Indian Penal Code 1860:
This act provides remedies and punishment in situations where an individual is harmed by another person's activity, and any person who violates the laws will be dealt with in accordance with the following provisions:

- **Sedition is covered in Section 124A.**
- **Section 153A addresses inciting hatred.**
- **Regarding the deliberate disrespect of religion, see Section 295A.**
- **Defamation is covered by Section 499.**
- **Public mischief is covered by Section 505**
- **Section 509: Disrespect for the modesty of women.**

Act of 1998 prohibiting indecent depictions of women Pornography is protected and prohibited by sections 3 and 4 of this Act, and those who distribute this kind of offensive

content are also subject to punishment. The indecent portrayal of women is punishable under Indian penal code sections 292 and 293.

BEST PRACTICES FOR THE USE OF SOCIAL MEDIA

Social networking sites like Facebook and Twitter may be a terrific way to stay in touch with friends. Here are some safety tips for social networking.
However, there are several safety precautions for social media that you should always follow.

- **Set your privacy preferences:** Utilise the privacy and security options on your social networking sites by becoming familiar with them. They assist you in managing your online experience well and in controlling who sees what you submit.
- **Recall that a post is permanent:** Ensure that your online reputation is protected. What you publish online remains there. Before uploading any images, you wouldn't want your parents or potential employers to view, give it some thought. According to a recent study, 70% of hiring managers disqualified candidates based only on information they discovered online.
- **Create a good online reputation:** Recruiters are more receptive to candidates with a strong, favorable internet reputation, according to recent research. So show off your skills and your command of the situation.
- **Keep private information private:** Be cautious when sharing too much personal information on social networking sites. More information you share online may make it simpler for someone to steal your identity, access your data, or engage in other crimes like stalking.
- **Safeguard your computer:** The first step in security is to secure your computer. Software antivirus installation. Keep your web browser, operating system, and other software up to date.
- **Know what to do:** Block the person, delete them from your list of friends, and report them to the site administrator if they are harassing or threatening you.
- **Make sure to use strong passwords:** Your password should be at least eight characters long and contain a mix of letters, numbers, and special characters (such as,@, #, or $).
- **Be careful when using social networking sites:** since occasionally even links that appear to be from friends include malicious software or are a part of a phishing campaign. Do not click it if you have any reason to be suspicious. Make contact with your friend to first confirm the legitimacy of the link.

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CASE STUDIES.

Case 1. TikTok's restriction on social media is the most recent scandal.
The Madras High Court issued an order in April 2019 telling the state government to forbid the use of the harmful TikTok app.
On June 29, 2020, the Indian government will outlaw the TikTok app, citing threats to the country's integrity, sovereignty, and public order. With the implementation of the 2009 Information Technology (Procedure and Safeguards for Blocking of Access to Information by Public) Rules, the act was outlawed under Section 69(a) of the Information and Technology Act.
The Ministry of Electronics and IT has received numerous complaints alleging that users' data is being stolen and moved outside of India in an unauthorized manner. The suggestions for preventing these apps from being used were made by the Indian Cyber Crime Coordination Centre.

Case 2. Related case law: In the 2015 case of Shreya Singhal v. Union of India, the Supreme Court decided that section 66A of the Information Technology Act violated the Constitution because it limited the freedom of speech and expression.

Case 3. It was determined in the case of Kharak Singh v. State of Uttar Pradesh that tapping phones violates privacy, and sharing information on WhatsApp with Facebook clearly breaches user privacy.

Case 4. In the Tehseen S. Poonawalla case from 2018, the Supreme Court granted the government complete discretion to halt the spread of explosive and reckless remarks on social media that have the potential to instigate any form of mob violence.

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UNIT - 4

Definition of E-Commerce

E-commerce (electronic commerce) is the activity of electronically buying or selling products on online services or over the Internet.
E-commerce draws on technologies such as mobile commerce, electronic funds transfer, supply chain management, Internet marketing, online transaction processing, electronic data interchange (EDI), inventory management systems, and automated data collection systems.

Main components of E-Commerce

1. **E-commerce platform**
   An e-commerce platform is a content management system (CMS) and commerce engine website used to manage catalogued products, register purchases, and manage a user relationship with an online retailer.
   Features of e-commerce platform.
   - Product catalog management
   - Shopping cart and checkout system
   - Payment gateway integration
   - Customer relationship management (CRM)
   - Order management and fulfillment
   - Inventory management

2. **Product Catalog**
   - A product catalog is a list of essential product details that help buyers make a purchase decision
   - These details include product features like descriptions, dimensions, price, weight, availability, colour, customer reviews, and more.

3. **Marketing**
   - Marketing is the process of promoting an online store and its products to potential customers.
   - This includes search engine optimization (SEO), pay-per-click (PPC) advertising, and social media marketing.

4. **Payment gateway**
   - A payment gateway authorizes the transfer of funds between buyers and sellers. It allows your e-commerce site to request money from a customer's bank for products or services that they have purchased.
   - Multiple online payment methods are one of the most significant components of the marketplace. Having multiple payment methods on your e-commerce store gives your customers the option to choose the preferred one. Also, online payments are quick and consume very little time.
   - Some platforms provide ‘‘buy now and pay later’’ option.
   - It provides secure payment processing

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5. **Security**
   E-commerce security is crucial for protecting sensitive customer data, preventing financial fraud, and maintaining website integrity. It encompasses various measures and practices to safeguard online transactions and ensure a secure shopping experience.

6. **Social networking**
   - A social network refers to an online community or platform where individuals can connect, interact, and engage with each other, as well as with businesses or brands.
   - Social networks play a significant role in the e-commerce landscape by providing a space for users to discover, share, and discuss products and services.

7. **Shipping and fulfilment**
   Order fulfillment involves storing products, packing these goods after customers place online orders, and then shipping these orders to customers.
   The major steps in the order fulfillment process include
   - Receiving products
   - Storing products
   - Picking, packing, and shipping products
   - Processing returns

8. **Customer service**
   Customer service is the process of assisting customers before, during, and after their purchase. This includes answering questions, resolving issues, and handling returns.
   - Multiple communication channels (phone, email, chat)
   - Self-service knowledge base
   - Live chat support

Elements of E-Commerce Security

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Confidentiality
Ensuring that sensitive customer Information should not be accessible to an unauthorized person, Such as credit card numbers and personal details. It should not be intercepted during the transmission. It involves

- Data encryption
- Access control
- Secure data storage

Integrity
Maintaining the accuracy and consistency of data throughout the e-commerce process, preventing unauthorized modifications or tampering with data. Data integrity ensures that the information processed and transmitted is reliable and trustworthy. It involves

- Data validation
- Data change tracking
- Data backups

Non-Repudiation

Establishing the authenticity of transactions and preventing any party from denying their involvement in a transaction. Digital signatures, timestamps, and audit trails can help establish non-repudiation.

- Digital signatures
- Timestamps
- Audit trails

Authenticity

- There should be a mechanism to authenticate a user before giving him/her access to the required information.
- Verifying the identities of users and entities involved in e-commerce transactions. Strong password policies, multi-factor authentication, and device fingerprinting can enhance authentication.

Privacy

- Privacy in e-commerce refers to the protection of personal data collected and processed by online retailers and other entities involved in the e-commerce ecosystem.
- It encompasses measures and practices to safeguard sensitive customer information, prevent unauthorized access and disclosure

Availability

Information should be available wherever and whenever required within a time limit specified.

- Ensuring that e-commerce websites and services are accessible to authorized users when needed. Robust infrastructure, DDoS protection, and disaster recovery plans are essential to maintain availability.

Encryption

- Encryption plays a crucial role in e-commerce by safeguarding sensitive customer data, preventing financial fraud, and maintaining website integrity.
- It ensures that confidential information exchanged between customers, merchants, and payment processors remains secure and protected from unauthorized access.

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E-Commerce threats

- E-commerce platforms and websites store sensitive customer data, such as credit card information, personal details, and purchase records. Hackers and cybercriminals can target these platforms to steal this data, leading to financial losses, identity theft, and reputational damage.
- Attackers intercept and alter communication between the customer and the e-commerce website. This can lead to the theft of sensitive information or the manipulation of transactions.
- Cross-Site Scripting (XSS): XSS attacks involve injecting malicious scripts into web pages viewed by other users. This can be exploited to steal session cookies or other sensitive information.

- Imposters can mirror your e-commerce site to steal customer money.
- Authorized administrators/users of an e-commerce website download hidden active content that attacks the e-commerce system.
- Some attackers develop special bots that can scrape your website to get information about inventory and prices. Such hackers, usually your competitors, can then use the data to lower or modify the prices in their websites in an attempt to lower your sales and revenue.
- Competitors or malicious actors may post fake reviews or manipulate ratings to harm a business's reputation or deceive customers.
- Insecure Application Programming Interfaces (APIs) can be exploited, leading to unauthorized access, data breaches, or manipulation of e-commerce processes.
- Without proper encryption, sensitive data transmitted between users and the e-commerce platform is at risk of interception by attackers.

E-commerce security best practices

- **Suggest strong passwords**
  Enforce strong password policies for both customers and employees. Encourage the use of complex passwords and two-factor authentication
- **Implement multilayer security**
- **Use a firewall**
- **Install a secure socket layer (SSL) certificate for HTTPS security**
  Implement SSL certificates to establish a secure connection and encrypt data. Regularly update and renew SSL certificates to maintain a strong security posture.
- **Follow payment card industry data security standard (PCI-DSS) requirements**
  If your business accepts credit card payments, adhere to PCI DSS standards. This set of security standards helps protect cardholder data and ensures secure payment transactions.
- **Data Encryption**
  Encrypt sensitive data, both in transit and at rest. This includes customer information, payment details, and any other confidential data stored on your servers
- **Secure Hosting and Server Configuration**
  Choose a reliable and secure hosting provider. Configure servers securely, and limit access to only necessary personnel. Implement firewalls and intrusion detection/prevention systems.
- **Implement Content Security Policies (CSP)**
  Utilize CSP to prevent cross-site scripting (XSS) attacks. This helps in controlling which resources can be loaded on your website and from where.
- **Backup Regularly:** Regularly back up your e-commerce website and databases. Ensure that backups are stored securely and test the restoration process periodically.
- **Update your software and hardware**
  Keep your e-commerce platform, content management system (CMS), plugins, and any other software up-to-date. Regularly apply security patches and updates to address vulnerabilities.

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Advantage of e-commerce

- E-commerce enables fast and secure shopping.
- Buying/selling a variety of goods and services from one's home or business anywhere, anytime transaction.
- E-commerce also enables you to choose different goods and services according to your choice.
- Can look for the lowest cost for specific goods or services.
- Consumers can easily compare prices and products from different retailers.
- Consumers can read product reviews and ratings from other customers before making a purchase.
- Consumers can receive personalized recommendations for products and services based on their past purchases and browsing behaviour.
- Businesses can reach out to worldwide clients can establish business partnerships.
- Supply chain management is simpler, faster, and cheaper using e-commerce.
  - Can order from several vendors and monitor supplies.
  - The production schedule and inventory of an organization can be inspected by a cooperating supplier who can in turn schedule their work.
- E-commerce replaced the paperwork as all transactions are through internet today.
- E-commerce is constantly evolving, with new technologies and innovations emerging all the time. This helps to keep the shopping experience fresh and exciting for consumers and businesses.

Survey of popular e-commerce sites.

E-commerce has revolutionized the way we shop, offering convenience, accessibility, and a vast array of products. With numerous platforms competing for customer attention, understanding their strengths and weaknesses is crucial for making informed purchasing decisions

Amazon:

Strengths:

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- Unrivaled product selection: Extensive inventory catering to diverse needs.
- Competitive pricing: Constant price wars ensure affordability.
- Streamlined logistics: Fast and reliable delivery across regions.
- Customer-centric approach: Focus on customer satisfaction and swift problem resolution.

Weaknesses:

- Overwhelming product selection: Navigation and decision-making can be challenging.
- Quality inconsistencies: Careful product selection is necessary due to varying quality levels.
- Ethical concerns: Facing criticism regarding labor practices and environmental impact.

Target Audience: Consumers seeking vast selection, competitive pricing, and efficient delivery.

EBay:

Strengths:

- Diverse marketplace: Offers new, used, and unique items, catering to diverse preferences.
- Competitive pricing: The bidding and negotiation format allows for lower prices.
- Community-driven atmosphere: Fosters interaction and personalized experiences.

Weaknesses:

- Product discovery challenges: Finding specific items can be difficult due to vast and unstructured inventory.
- Seller reliability considerations: Careful evaluation of sellers is required due to potential inconsistencies.
- Buyer shipping responsibility: Shipping costs can add to the overall price.

Target Audience: Value-conscious shoppers seeking unique items and a community-driven experience.

AliExpress:

Strengths:

- Exceptional affordability: Extremely low prices attract budget-conscious shoppers.
- Free shipping options: Reduced costs for many items.
- Global product assortment: A diverse range of products from China caters to international markets.

Weaknesses:

- Shipping delays: Slow delivery times from China.
- Quality variations: Potential for inconsistent product quality.
- Language barriers: Communication with sellers can be challenging.

Target Audience: Budget-conscious consumers seeking affordability and global product access.

Walmart:

Strengths:

- Convenient physical locations: Easy access to products and services.
- Competitive pricing: Affordable offerings across a broad range of products.
- One-stop shopping experience: Diverse product selection catering to diverse household needs.

Weaknesses:

- Limited online product availability: Not all store products are readily available online.
- Website navigation issues: Navigation can be slow and challenging.
- Labor practices scrutiny: Facing criticism regarding its treatment of workers.

Target Audience: Value-conscious consumers seeking convenience, affordability, and a one-stop shopping experience.

Etsy

Strengths:

- Unique handcrafted items: Offers personalized and one-of-a-kind products.
- Supports small businesses: Empowers independent artists and entrepreneurs.
- User-friendly platform: Easy navigation and a pleasant shopping experience.

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Weaknesses:

- Higher price points: Handcrafted nature leads to higher prices compared to mass-produced items.
- Quality variations: Product quality may vary depending on the seller's skills.
- Shipping costs considerations: Shipping costs can add to the overall price of handcrafted items.

Target Audience: Consumers seeking unique handcrafted items and supporting small businesses.

INTRODUCTION TO DIGITAL PAYMENTS

Digital payments are transactions that take place via digital or online modes, with no physical exchange of money involved. This means that both parties, the payer and the payee, use electronic mediums to exchange money.

Digital payments happen through various methods

Mobile wallets

Mobile Wallets, as the name suggests, are a type of wallet in which you can carry cash but in a digital format. Often customers link their bank accounts or banking cards to the wallet to facilitate secure digital transactions. Another way to use wallets is to add money to the Mobile Wallet and use the said balance to transfer money.

Unstructured Supplementary Service Data (USSD)

USSD was launched for those sections of India's population that don't have access to proper banking and internet facilities. Under USSD, mobile banking transactions are possible without an internet connection by simply dialing *99# on any essential feature phone.

Aadhaar Enabled Payment System (AEPS)

AEPS is a bank-led model for digital payments that was initiated to leverage the presence and reach of Aadhaar. Under this system, customers can use their Aadhaar-linked accounts to transfer money between two Aadhaar-linked Bank Accounts.

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Unified Payments Interface (UPI)

UPI is a payment system that culminates numerous bank accounts into a single application, allowing the transfer of money easily between any two parties. As compared to NEFT, RTGS, and IMPS, UPI is far more well-defined and standardized across banks. You can use UPI to initiate a bank transfer from anywhere in just a few clicks.

POS Terminals

POS (Point of Sale) is known as the location or segment where a sale happens. For a long time, POS terminals were considered to be the checkout counters in malls and stores where the payment was made. The most common type of POS machine is for Debit and Credit cards, where customers can make payment by simply swiping the card and entering the PIN.

Internet Banking

Internet Banking, also known as e-banking or online banking, allows the customers of a particular bank to make transactions and conduct other financial activities via the bank's website. E-banking requires a steady internet connection to make or receive payments and access a bank's website, which is called Internet Banking.

Mobile Banking

Mobile banking refers to the act of conducting transactions and other banking activities via mobile devices, typically through the bank's mobile app. Today, most banks have their mobile banking apps that can be used on handheld devices like mobile phones and tablets and sometimes on computers.

Benefits of digital payment

- Cash management is eliminated resulting in a lower risk of theft and reducing the cost of security and storage.
- Digital payments are often quicker transactions, thereby resulting in shorter queues and enhancing the customer's in-shop experience. Customer convenience is thus driving sales.
- A clear trail is available for easy accounting, helping simplify operations and tax compliance.
- Safety and efficient tracking
- Global reach, send, and receive money across borders effortlessly.
- Real-time updates, get instant confirmation of your transactions.

Modes of digital payments

Banking Cards

- Physically, a bank card will usually have the client's name, the issuer's name, and a unique card number printed on it.
- It will have a magnetic strip on the back enabling various machines to read and access information.

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- this may allow the card to be used as an ATM card, enabling transactions at automated teller machines or as a debit card, linked to the client's bank account and able to be used for making purchases at the point of sale with a bank card using a payment terminal.
- Banking cards offer consumers more security, convenience, and control than any other payment method.
- The wide variety of cards available – including credit, debit, and prepaid – offers enormous flexibility. These cards provide 2-factor authentication for secure payments e.g. secure PIN and OTP.
- RuPay, Visa, and MasterCard are some of the examples of card payment systems.
- Payment cards give people the power to purchase items in stores, on the Internet, through mail-order catalogs, and over the telephone.
- They save both customers and merchants time and money and thus enable them for ease of transaction.

Types of Banking cards

- Debit card:- Debit cards are linked to the holder's bank account and allow them to spend money directly from their available funds. Transactions are deducted from the account balance in real time.
- Credit card:- Issued by banks, credit cards provide a line of credit to users. Cardholders can make purchases up to a predefined credit limit. With interest charged on unpaid amounts.
- Prepaid cards:- These cards are loaded with a predetermined amount of money They can be used for transactions until the balance reaches zero.
- ATM cards:- Primarily designed for cash withdrawals from ATMs.

How to do transactions with Bank cards

- POS terminal or online payment gateway
- Present Card physically or card details for online transaction
- Provide PIN
- Provide OTP (One Time Password) received on registered mobile to complete online transactions for merchant website.

Transaction Cost

- NIL to the customer for merchant transactions.
- Annual fee and limits on ATM transactions by banks discretion.
- 0.50% to 2.25% paid by merchant
- Cash-out charged to the customer in case of credit cards @ 1% to 3.5% of the transaction value.

Services Offered

- These cards can be used at PoS (Point of Sale) machines, ATMs, micro ATMs, Shops, wallets, online transactions, and e-commerce websites.
- International cards can be used across the globe for multiple currencies

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Unified Payment Interface (UPI)

UPI, or Unified Payments Interface, is an instant real-time payment system developed by the National Payments Corporation of India (NPCI). It facilitates inter-bank peer-to-peer (P2P) transactions. This means that you can transfer money between any two bank accounts in India, as long as they both have a UPI ID.

How does UPI work?

- UPI transactions are initiated and completed using a mobile phone application.
- To use UPI, you will need to download a UPI app from your bank or a third-party provider. Once you have downloaded the app, you will need to link your bank account to it.
- To make a UPI payment, you will need to enter the recipient's phone number or virtual payment address (VPA).
- Then need to enter the amount of money you want to send and confirm the transaction.

- The transaction will be processed instantly, and the recipient will receive the funds in their bank account within seconds.

Here are some of the key features of UPI:

- **Real-time payments:** UPI transactions are processed instantly, so you can send or receive money within seconds.
- **Single mobile application:** You can link multiple bank accounts to a single UPI app, so you don't need to switch between apps to make payments from different accounts.
- **Single Virtual Payment Address (VPA):** Each UPI user is assigned a unique identifier called a VPA. This VPA is like your phone number for payments - you can share it with others to receive money.
- **Multiple payment options:** UPI supports a variety of payment options, including P2P transfers, bill payments, merchant payments, and online purchases.
- **Security:** UPI employs robust security measures, such as two-factor authentication (2FA) and encryption, to protect user data and transactions.

How UPI is unique from other payment modes

- Immediate money transfer through mobile device round the clock 24*7 and 365 days.
- Single mobile application for accessing different bank accounts.
- Single Click 2 Factor Authentication – Aligned with the Regulatory guidelines, yet provides for a very strong feature of seamless single click payment.
- Virtual address of the customer for Pull & Push provides for incremental security with the customer not required to enter the details such as Card no, Account number; IFSC etc.
- QR Code
- Best answer to Cash on Delivery hassle, running to an ATM or rendering exact amount.
- Merchant Payment with Single Application or In-App Payments.
- Utility Bill Payments, Over the Counter Payments, QR Code (Scan and Pay) based payments.
- Donations, Collections, Disbursements Scalable.
- Raising Complaint from Mobile App directly.

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UPI has revolutionized the digital payments landscape in India, driving significant growth in cashless transactions. Its popularity has contributed to financial inclusion and increased accessibility to financial services.

E-Wallets

An electronic wallet, commonly known as an e-wallet, is a digital tool that allows individuals to store, manage, and transact money electronically.

An e-wallet is a digital application that allows to make electronic transactions using their mobile devices or computers. It securely stores payment information, such as

credit card numbers and bank account details, eliminating the need to carry physical cards.

This can include purchasing items either online or at the point of sale in a store, using either mobile payment or (for online buying only) using a laptop or other personal computer

- Digital wallets are stored on the client side and are easily self-maintained and fully compatible with most e-commerce websites.
- Digital wallets are composed of both digital wallet devices and digital wallet systems. There are dedicated digital wallet devices such as the biometric wallet by Dunhill, a physical device that holds cash and cards along with a Bluetooth mobile connection.
- Presently there are further explorations for smartphones with NFC digital wallet capabilities, such as smartphones utilizing Google's Android and Apple's iOS operating systems to power wallets such as Google Pay and Apple Pay.
- Digital wallet systems enable the widespread use of digital wallet transactions among various retail vendors in the form of mobile payment systems and digital wallet applications.
- Digital wallets are being used more frequently among Asian countries.
- Indian consumers are leading the way with 76.4% using a smartphone to make a purchase.
- Companies like Reliance and Amazon India to come out with their digital wallet. Flipkart has already introduced its digital wallet.

Here are some types of e-wallets:

- **Closed wallets:**- Can only be used for transactions within the company's ecosystem. **Example:**- Amazon wallet.
- **Semi-closed wallets**:- Can be used for shopping and transferring funds to other users in the same network. **Example:-** Paytm wallet
- **Open wallets:**- Can be used directly by a bank or through a third party. **Example:**- Google pay, Apple pay.
- **Crypto wallets**:- Can be used to send, receive, and store cryptocurrencies. **Example:-** Electrum, MetaMask.

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Benefits of E-wallet

- **Convenience**:- E-wallets allow users to store and access multiple payment methods in one place, eliminating the need to carry physical cards or cash. They can be accessed through mobile devices, making payments fast and convenient.
- **Security:-** Most, if not all digital wallets offer advanced security features e.g. biometric authentication and encryption, this protects the financial information of the users thus preventing fraud.
- **Cost-effective:-** E-wallets can be more cost-effective than traditional payment methods, as they often offer lower transaction fees and can reduce the need for physical infrastructure such as card readers.
- **Better record-keeping:-** E-wallets provide users with a detailed record of their transactions, making it easier to make follow-up on spending and manage budgets.
- **Global acceptance:**- E-wallets are widely accepted by merchants worldwide, both online and in-store.

Unstructured Supplementary Service Data (USSD)

Unstructured Supplementary Service Data (USSD) is a communication protocol used by GSM (Global system for mobile communication) cellular telephones to communicate with the mobile network operator's computers. It is a text-based protocol that allows users to access and interact with a variety of services, such as checking their account balances, transferring money, and activating new services.

How does USSD work?

USSD works by sending a text message from the user's phone to the mobile network operator. The message contains a code, also known as a USSD string, that identifies the service the user wants to access. The mobile network operator's computer then processes the code and sends a response back to the user's phone. The response can be text, such as a menu of options, or data, such as account balance information.

Functionality

- **Initiating and completing transactions:** USSD allows users to send and receive money, pay bills, and recharge accounts without needing an internet connection. This is particularly helpful in rural areas or for people with limited access to smartphones or data plans.
- **Checking account balance and transaction history:** Users can access their account information and transaction history through USSD menus, providing them with real-time insights into their financial activity.
- **Activating and managing services:** USSD facilitates the activation and deactivation of various services linked to digital payments, such as mobile wallets and e-banking features.
- **Providing customer support:** Banks and financial institutions can utilize USSD to offer basic customer support services, like answering common questions and providing instructions.

Benefits of USSD

- **Simplicity:** USSD is a very simple protocol to use. Users do not need to have any special knowledge or equipment to use USSD.
- **Speed:** USSD transactions are processed very quickly. Users can typically get a response within seconds.
- **Cost-effectiveness:** USSD transactions are typically free or have minimal charges.
- **Accessibility:** USSD is accessible on all GSM phones, regardless of the phone's model or features.

Limitations

- **Limited functionality:** Compared to other digital payment methods, USSD offers a limited range of features and functionalities.
- **Limited data transmission:** USSD can only handle text-based data, restricting its ability to transmit complex information or large files.
- **User experience:** The user interface of USSD menus can be text-heavy and less intuitive compared to graphical interfaces.

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- **Security concerns:** While USSD offers some security measures, it is important for users to be aware of potential phishing scams and protect their confidential information.

AADHAR ENABLED PAYMENTS

Aadhaar Enabled Payment System (AEPS) is a system that allows Aadhaar cardholders to make financial transactions using their Aadhaar number and biometric authentication. AEPS was introduced by the National Payments Corporation of India (NPCI) to promote digital financial inclusion and make banking services accessible to all sections of the society

- Banking Services Offered by AePS
- Cash Deposit
- Cash Withdrawal
- Balance Enquiry
- Mini Statement
- Aadhaar to Aadhaar Fund Transfer
- Authentication
- BHIM Aadhaar Pay

How it works

- Link your Aadhaar to a bank account: Ensure your 12-digit Aadhaar number is linked to your active bank account for streamlined transactions.
- Locate an AEPS kiosk: Look for micro ATMs or shops displaying the "AEPS" sign. They're widely available across India, especially in rural areas.
- Authenticate with your fingerprint: Place your finger on the biometric sensor to verify your identity. No need for cards or PINs!
- Choose your transaction: Select the desired service from options like cash withdrawal, balance inquiry, fund transfer, or even bill payments.
- Enter the amount (if applicable): Key in the desired amount for cash withdrawal or fund transfer.
- Confirm and complete: Review the transaction details and authorize it with your fingerprint again.

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Benefits of AEPS

- **Convenience:** Access your bank account instantly, even without a debit card or smartphone.
- **Security:** Biometric authentication eliminates the risk of card theft or PIN fraud.
- Financial inclusion: Enables financial services in remote areas for underbanked populations.
- **Affordability:** Transaction fees are typically lower compared to traditional channels.
- **Transparency:** Real-time updates on your bank account balance after each transaction.

DIGITAL PAYMENTS RELATED COMMON FRAUDS AND PREVENTIVE MEASURES.

Digital payments related common frauds and preventive measures

1. Phishing

- **Fraud Description:** Users receive fake emails, messages, or calls that mimic legitimate sources, tricking individuals into providing sensitive information.
- **Preventive Measures**
  - Verify the legitimacy of communication channels.
  - Never share personal or financial information through unsolicited emails or messages.
  - Educate users on recognizing phishing attempts.

2. Identity Theft

- **Fraud Description:** Criminals steal personal information to impersonate the victim, opening accounts or making transactions in their name.
- **Preventive Measures:**
  - Use strong, unique passwords for all accounts.
  - Enable two-factor authentication (2FA) where available.
  - Regularly monitor credit reports for any suspicious activity.

3. Card Skimming

- **Fraud Description:** Devices capture card information during legitimate transactions, often at ATMs or point-of-sale terminals.
- **Preventive Measures:**
  - Inspect card readers for any irregularities.
  - Shield PIN entry when making transactions.
  - Regularly check bank statements for unauthorized transactions.

4. Man-in-the-Middle Attacks

- **Fraud Description:** Hackers intercept and alter communication between parties during a transaction.
- **Preventive Measures:**
  - Use secure, encrypted connections (HTTPS).
  - Avoid public Wi-Fi for sensitive transactions.
  - Employ secure communication channels.

5. Unauthorized Transactions:

- **Fraud Description:** Criminals gain access to a user's account and make transactions without their knowledge.
- **Preventive Measures:**
  - Regularly change passwords and use strong authentication methods.
  - Enable transaction alerts for unusual activity.
  - Implement biometric authentication where available.

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6. Mobile Wallet Frauds

- **Fraud Description:** Criminals compromise mobile wallets to gain unauthorized access and make transactions.
- **Preventive Measures:**
  - Keep mobile devices and wallet apps updated.
  - Use security features such as PIN or biometric authentication.
  - Enable app lock features.

7. Vishing (Voice Phishing)

- Fraud Description: Attackers use phone calls to trick individuals into providing personal or financial information.
- **Preventive Measures:**
  - Be skeptical of unexpected calls asking for sensitive information.
  - Verify the caller's identity through official channels.
  - Educate users on recognizing vishing attempts.

8. Social Engineering

- **Fraud Description:** Manipulating individuals into divulging confidential information or performing actions against their best interest.
- **Preventive Measures:**
  - Train users to recognize social engineering tactics.
  - Implement policies to verify identity before sharing sensitive information.
  - Be cautious about sharing personal details on social media.

9. Fake QR codes: Malicious QR codes can redirect you to phishing websites or download malware onto your device.

RBI GUIDELINES ON DIGITAL PAYMENTS AND CUSTOMER PROTECTION IN UNAUTHORIZED BANKING TRANSACTIONS.

An unauthorised transaction is when someone transfers money from your account without your permission.

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RBI Guidelines for Unauthorised Transactions

If neither the bank nor the customer is at fault and a third-party breach led to an unauthorised transaction complaint, the following rule applies for a refund:

- Within 3 Days - As per RBI guidelines for disputed transactions, if the customer reports bank fraud transactions within 3 days, he/ she will have to bear zero liability. The whole amount will be refunded to the account by the concerned bank.
- Within 4-7 Days - Limiting liability of customers, RBI guidelines require customers to bear with the transaction amount or an amount within Rs 5000 up to Rs 25,000, whichever is lower in case the unauthorised transaction complaint is made after 3 days and before 7 days.
- After 7 Days - The bank won't refund unauthorised transactions after the expiry of 7 days if the policy does not allow it.
- In cases where the loss is due to negligence by a customer, such as where he has shared the payment credentials, the customer will bear the entire loss until he reports

the unauthorised transaction to the bank. Any loss occurring after the reporting of the unauthorised transaction shall be borne by the bank.

Documents Required for Unauthorised Transaction Complaint

- Bank Account Statement
- Address Proof of Account Owner
- Identity Proof
- Xerox copies of unauthorised transaction snapshots
- Online FIR for fraud transaction at Cyber Crime website
- A written complaint containing complete details of the unauthorised transaction on the credit card or other mode

CUSTOMER PROTECTION IN UNAUTHORIZED BANKING TRANSACTIONS.

The systems and procedures in banks must be designed to make customers feel safe about carrying out electronic banking transactions. To achieve this, banks must put in place:

- appropriate systems and procedures to ensure safety and security of electronic banking transactions carried out by customers;
- robust and dynamic fraud detection and prevention mechanism;
- mechanism to assess the risks (for example, gaps in the bank's existing systems) resulting from unauthorised transactions and measure the liabilities arising out of such events;
- appropriate measures to mitigate the risks and protect themselves against the liabilities arising therefrom; and
- a system of continually and repeatedly advising customers on how to protect themselves from electronic banking and payments related fraud.
- The RBI actively promotes cyber awareness campaigns to educate customers about safe digital payment practices and common fraud tactics.
- Banks are mandated to implement robust cybersecurity measures to protect customer data and infrastructure from cyberattacks.

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RELEVANT PROVISIONS OF PAYMENT SETTLEMENT ACT,2007.

The Reserve Bank of India (RBI) established the Payment and Settlement Act System, 2007 (PSS Act, 2007), which was approved by the President on December 20, 2007. It becomes effective on August 12, 2008. The Board for Regulation and Supervision of Payment and Settlement Systems is a central authority that has been established by RBI

Authorization and Regulation:

- Section 4: No payment system can operate without authorization from the Reserve Bank of India (RBI), ensuring only certified and compliant systems handle your transactions.
- Section 10: RBI sets standards for security, risk management, and consumer protection that payment systems must adhere to. This ensures consistent safety across different platforms.

Dispute Resolution:

- Section 24: A dedicated mechanism for settling disputes arising from unauthorized or erroneous transactions. Customers can escalate complaints through their bank, the RBI Ombudsman, or even courts.
- Zero Liability Framework: Under certain conditions (timely reporting, no negligence), customers are not liable for unauthorized transactions, safeguarding their finances.

Information Security:

- Section 13: Provides access to information for investigating potential threats and unauthorized activities, leading to faster resolution of disputes.
- Section 16: Empowers RBI to conduct audits and inspections of payment systems, ensuring adherence to security and data protection protocols.

Information Security:

- Section 13: Provides access to information for investigating potential threats and unauthorized activities, leading to faster resolution of disputes.
- Section 16: Empowers RBI to conduct audits and inspections of payment systems, ensuring adherence to security and data protection protocols.

Consumer Protection:

- Section 25: Outlines specific requirements for electronic funds transfers, including clear disclaimers and timely notifications, reducing confusion and potential misuse.
- Section 26 and 27: Define penalties for non-compliance and offenses related to payment systems, holding violators accountable and deterring fraudulent activities.

Additional Provisions:

- Section 32: The Act supersedes other laws regarding payment systems, ensuring consistent and comprehensive customer protection across India.
- Section 34: Excludes stock exchanges and their clearing corporations from the Act's direct application, focusing on retail and consumer-oriented payment systems.

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UNIT - V

END POINT DEVICE AND MOBILE PHONE SECURITY

END POINT DEVICES SECURITY

Endpoint security refers to the practices and technologies employed to secure network endpoints, including laptops, desktops, servers, and mobile devices. It encompasses a range of measures, such as antivirus software, firewalls, encryption, and access controls, to protect endpoints from unauthorized access and malicious activities.

- Install reputable antivirus and anti-malware software on endpoint devices. Keep the software updated to defend against the latest threats.
- Regularly update the operating system and software applications to patch security vulnerabilities. Enable automatic updates when possible.
- Enable firewalls on endpoint devices to monitor and control incoming and outgoing network traffic. This helps prevent unauthorized access and protects against malware.
- Use encryption for sensitive data, both at rest and in transit. This ensures that even if the device is compromised, the data remains unreadable without the proper decryption key.
- Implement strong password policies and consider multi-factor authentication (MFA) for an additional layer of security. This helps prevent unauthorized access to devices.
- Consider using Endpoint Detection and Response solutions to detect and respond to security incidents on endpoint devices in real-time.
- Regularly back up important data to an external, secure location. This ensures that data can be restored in case of a device failure or a security incident.
- Implement device management solutions to monitor and control the configuration and security of endpoint devices within an organization.

MOBILE PHONE SECURITY

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Mobile devices face unique risks, including app-based vulnerabilities, mobile malware, and data leakage. With the growing reliance on mobile devices for work-related tasks, securing these endpoints becomes paramount. Organizations must adopt a holistic approach to mobile device protection.

- Use strong passwords, PINs, or biometric authentication (fingerprint, face recognition) to secure access to mobile devices.
- Review and manage app permissions. Only grant necessary permissions to applications, and be cautious about granting access to sensitive data.
- Download and install apps only from official app stores to reduce the risk of downloading malicious software.
- Keep the mobile operating system and apps up-to-date with the latest security patches. Enable automatic updates when possible.
- Enable remote wiping capabilities to erase data in case the mobile device is lost or stolen.

- Avoid connecting to unsecured Wi-Fi networks. Use Virtual Private Network (VPN) services when accessing sensitive information over public Wi-Fi.
- For enterprise environments, consider using Mobile Device Management solutions to enforce security policies and remotely manage devices.
- Install security apps that provide features such as anti-malware, remote tracking, and data protection.
- Educate users about mobile security best practices, including the risks of downloading unknown apps, clicking on suspicious links, and the importance of keeping their devices secure.

PASSWORD POLICY

A password policy is a set of rules designed to enhance computer security by encouraging users to employ strong passwords and use them properly. A password policy is often part of an organization's official regulations and may be taught as part of security awareness training.

Password length and formation

Many policies require a minimum password length. Eight characters is typical but may not be appropriate. Longer passwords are generally more secure

Some policies suggest or impose requirements on what type of password a user can choose, such as:

- The use of both upper-case and lower-case letters (case sensitivity).
- Inclusion of one or more numerical digits.
- Inclusion of special characters, such as @, #, $.
- Prohibition of words found in a password blocklist.
- Prohibition of words found in the user's personal information.
- Prohibition of use of company name or an abbreviation.
- Prohibition of passwords that match the format of calendar dates, license plate numbers, telephone numbers, or other common numbers.

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The key points

- Verifiers should not impose composition rules e.g., requiring mixtures of different character types or prohibiting consecutively repeated characters
- Verifiers should not require passwords to be changed arbitrarily or regularly e.g. the previous 90 day rule
- Passwords must be at least 8 characters in length
- Password systems should permit subscriber-chosen passwords at least 64 characters in length.
- All printing ASCII characters, the space character, and Unicode characters should be acceptable in passwords
- When establishing or changing passwords, the verifier shall advise the subscriber that they need to select a different password if they have chosen a weak or compromised password

- Verifiers should offer guidance such as a password-strength meter, to assist the user in choosing a strong password
- Verifiers shall store passwords in a form that is resistant to offline attacks. Passwords shall be salted and hashed using a suitable one-way key derivation function.

SECURITY PATCH MANAGEMENT

Patch management is the process of applying updates to software, drivers, and firmware to protect against vulnerabilities. Effective patch management also helps ensure the best operating performance of systems, boosting productivity.

- Patch management is the practice of deploying firmware, driver, operating system (OS), and application updates to your computing endpoints.
- Patch management is critical to keeping systems updated, reducing attack surfaces, and ensuring employee productivity.
- Unpatched vulnerabilities are open doors for attackers, allowing them to exploit weaknesses and steal data, disrupt systems, or unleash malware.
- Hackers and cybercriminals constantly devise new exploits, so proactive patching keeps you one step ahead, closing known vulnerabilities before they can be weaponized.
- Track and identify available patches for your operating systems, applications, and firmware. Utilize automated tools to streamline the process.
- Before deploying patches on critical systems, test them in a controlled environment to ensure compatibility and avoid unforeseen disruptions.
- Track patch deployment progress and monitor systems for any adverse effects.

Benefits of Effective Patch Management:

- **Reduced security risks**: Mitigate vulnerabilities that could be exploited by attackers.
- **Improved system uptime:** Prevent downtime caused by malware infections or system crashes.
- **Enhanced compliance:** Meet regulatory requirements and industry standards.
- **Optimized IT resources:** Streamline patch management processes and reduce IT workload.

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DATA BACKUP

Data backup is the process of creating copies of your valuable digital data and storing them safely in a separate location. It serves as a safety net, ensuring you can recover your data in case of primary data loss due to hardware failures, software corruption, accidental deletion, cyberattacks, or natural disasters.

Key Concepts in Data Backup

- **Backup Location:** Where you store your backups. Common options include:
  - External hard drives
  - Cloud storage services

- Network-attached storage (NAS) devices
- Backup servers

- **Backup Frequency:** How often you create backups.
  - **Full backups:** Copy all data every time.
  - **Incremental backups:** Copy only new or changed data since the last backup.
  - **Differential backups:** Copy all data changed since the last full backup.
  - **Backup Retention:** How long you keep backups before overwriting them.
  - **Backup Restoration:** Restoring lost or damaged data from a backup.

- **Automate Backups:**
  Automate the backup process to ensure consistency and reliability. Scheduled automated backups reduce the risk of human error.

- **Encryption:**
  Encrypt sensitive data during transit and storage to ensure data confidentiality. This is especially crucial for offsite or cloud backups.

Benefits of Data Backup

- **Data Protection:** Safeguards against data loss and minimizes downtime.
- **Business Continuity:** Ensures operations can continue even after data loss incidents.
- **Compliance:** Meets regulatory requirements for data retention and disaster recovery.
- **Peace of Mind:** Provides assurance that your data is protected.

Best Practices for Data Backup

- **Create a backup plan:** Define what data to back up, how often, and where to store backups.
- **Use a reliable backup solution**: Choose a reputable software or service with features like encryption, compression, and scheduling.
- **Test your backups regularly**: Ensure backups are working correctly and can be restored.
- **Store backups securely:** Protect backups from unauthorized access, theft, or destruction.

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DOWNLOADING AND MANAGEMENT OF THIRD-PARTY SOFTWARE

Third-party software is a computer program created or developed by a different company than the one that developed the computer's operating system

Downloading third-party software

- Only download software from reputable sources, preferably the official websites of the developers or trusted online stores. Avoid suspicious websites and third-party download platforms.

- Before downloading, check user reviews and independent tests to get insights into the software's functionality, security, and overall reputation.
- Always review the permissions requested by the software during installation. Be wary of programs asking for excessive access to your system or data.
- Utilize reputable antivirus and anti-malware software to scan downloaded files before installing them. This helps detect and prevent potential threats.
- Use endpoint protection solutions, including antivirus and anti-malware software, to scan and monitor downloaded files for potential threats

Management of third-party software

- Regularly update installed software to benefit from security patches and bug fixes. Enable automatic updates whenever possible to simplify the process.
- Avoid installing unnecessary software. The more programs you have, the higher the security risk and the larger the drain on your system resources.
- Periodically review your installed programs and remove any you no longer use. This reduces potential vulnerabilities and clutter.
- Use system monitoring tools to identify programs running in the background and consuming resources. This helps identify potentially unwanted applications.
- Occasionally review the permissions granted to installed programs and revoke any that are no longer necessary.

DEVICE SECURITY POLICY

Device security is the process of securing mobile phones, laptops, PCs, and Internet of Things (IoT) devices from cyber threats and unauthorized access.

Device security is important for safeguarding sensitive data and helping to prevent unauthorized access, data breaches, and other threats. Securing devices connected to corporate networks can help mitigate risks

To secure your devices, a multilayered approach is required. Start by training users in security best practices. Implement a zero trust network access (ZTNA) policy to mitigate the risk of unauthorized access. Deploy device security tools to fortify defenses against threats and safeguard the network.

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IoT device security helps prevent IoT devices from introducing threats to a network. IoT devices include any network-connected hardware. Smart thermostats, security cameras, printers, and industrial equipment are all examples of IoT devices.

1. User Authentication

Define requirements for user authentication, including strong password policies, multi-factor authentication (MFA), and biometric authentication where applicable.

2. Access Controls

Specify access controls and permissions to restrict users' access to data and resources based on their roles and responsibilities. Implement the principle of least privilege.

3. Encryption

Mandate the use of encryption for data both at rest and in transit. This includes disk encryption, encrypted communication protocols (e.g., TLS), and encryption of sensitive files.

4. Device Configuration Standards

Establish standardized configurations for devices, including operating system settings, application configurations, and security settings. Regularly update and patch devices to address vulnerabilities.

5. Endpoint Protection

Require the installation and regular updating of endpoint protection solutions, such as antivirus, anti-malware, and endpoint detection and response (EDR) tools.

6. Security Patch Management

Define procedures for timely and regular application of security patches and updates to address vulnerabilities and ensure that devices are running the latest, secure software versions.

7. Physical Security

Implement measures to physically secure devices, such as access controls to server rooms, laptop lock mechanisms, and policies for reporting lost or stolen devices.

CYBER SECURITY BEST PRACTICES

Cybersecurity best practices are essential for individuals, organizations, and businesses to protect their digital assets and sensitive information from cyber threats.

Here is a comprehensive list of cybersecurity best practices:

For Individuals

1. **Use Strong Passwords:**
   Create complex passwords and use a combination of uppercase and lowercase letters, numbers, and special characters. Avoid easily guessable information like birthdays or common words.
2. **Enable Multi-Factor Authentication (MFA):**
   Enable MFA whenever possible to add an extra layer of security to your accounts.
3. **Regularly Update Software:**
   Keep operating systems, applications, and antivirus software up to date with the latest security patches.
4. **Be Cautious with Email:**
   Avoid clicking on suspicious links or downloading attachments from unknown sources. Be cautious about phishing emails.
5. **Use Secure Wi-Fi Connections:**
   Secure your home Wi-Fi with a strong password and encryption. Avoid using public Wi-Fi for sensitive transactions.
6. **Regularly Back Up Data:**
   Back up important data regularly to an external device or cloud service. This helps in case of data loss due to ransomware or hardware failure.
7. **Keep Personal Information Private:**
   Be mindful of the information you share online. Avoid oversharing on social media platforms and use privacy settings.

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8. **Use a Firewall:**
   Enable the firewall on your devices to control incoming and outgoing network traffic.
9. **Secure Your Devices:**
   Use device passwords or biometrics to secure smartphones, laptops, and other devices. Implement remote tracking and wiping features.
10. **Stay Informed:**
    Keep yourself updated on the latest cybersecurity threats and best practices. Follow reputable cybersecurity news sources.

For Organizations

1. **Develop a Cybersecurity Policy:**
   Create and enforce a comprehensive cybersecurity policy that includes user guidelines, access controls, and incident response procedures.
2. **Employee Training:**
   Provide regular cybersecurity awareness training for employees to recognize and respond to potential threats.
3. **Network Security:**
   Implement firewalls, intrusion detection systems, and other network security measures to protect against unauthorized access.
4. **Regularly Update and Patch Systems:**
   Keep all software, including operating systems and applications, updated with the latest security patches.
5. **Incident Response Plan:**
   Develop and regularly test an incident response plan to efficiently handle and recover from cybersecurity incidents.
6. **Regular Security Audits:**
   Conduct regular security audits and vulnerability assessments to identify and address potential weaknesses.
7. **Backup and Recovery:**
   Establish regular data backup procedures and test data recovery processes to ensure business continuity in case of data loss.
8. **Secure Third-Party Relationships:**
   Evaluate the cybersecurity practices of third-party vendors and service providers. Ensure they meet your security standards.
9. **Endpoint Security:**
   Implement endpoint protection measures, including antivirus software, on all devices connected to the organization's network.
10. **Security Awareness Training:**
    Train employees on cybersecurity best practices, including how to recognize and report security incidents.
11. **Regularly Review and Update Policies:**
    Periodically review and update cybersecurity policies to adapt to evolving threats and technological changes.

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SIGNIFICANCE OF HOST FIREWALL AND ANT-VIRUS

SIGNIFICANCE OF HOST FIREWALL

A host-based firewall is firewall software that is installed directly on a computer (rather than a network). Host-based firewalls help detect and stop viruses, malware and other malicious scripts that may not have been caught by network security.

- **Network Security**
  The host firewall acts as a barrier between the device and the network, controlling incoming and outgoing network traffic based on predetermined security rules. It helps prevent unauthorized access to the device and protects against network-based attacks.
- **Protection Against Unauthorized Access**
  By filtering incoming and outgoing traffic, a host firewall prevents unauthorized entities from accessing the device. This is crucial for safeguarding sensitive data and ensuring the privacy of the user.
- **Defense Against Malware**
  Host firewalls can block incoming malicious traffic and prevent malware from establishing unauthorized connections. This is particularly important in protecting against malware that may attempt to communicate with external servers.
- **Security on Public Networks**
  When connected to public Wi-Fi or other untrusted networks, a host firewall adds an extra layer of defense by blocking potentially malicious incoming connections. This is especially important for users on the go.
- **Mitigation of Exploits**
  Firewalls can block certain types of network traffic associated with exploits, helping to protect against vulnerabilities in software and operating systems.

SIGNIFICANCE OF ANT-VIRUS

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An antivirus (or anti-virus) is a type of software designed to detect, prevent, and remove malicious software (malware) from computer systems. The primary goal of antivirus software is to protect devices from various types of threats, including viruses, worms, trojans, spyware, ransomware, and other forms of malicious code.

How antivirus keeps your digital city safe:

- **Detection:** Like skilled detectives, antivirus programs constantly scan your files, emails, and downloads for suspicious activity. They utilize signature-based and heuristic methods to identify known and unknown threats, ensuring no rogue agent slips through the cracks.
- **Neutralization:** Once a threat is identified, antivirus acts swiftly to neutralize it. This might involve quarantining the infected file to prevent further harm, removing it entirely, or even repairing damaged files. Imagine these agents swiftly apprehending and disarming the criminals before they can cause chaos.
- **Prevention:** But the best defense is a good offense. Antivirus software uses various methods, like blocking malicious websites and downloads, to prevent threats from

entering your system in the first place. Think of this as fortified city walls and vigilant guards at the gates, stopping troublemakers before they even reach the doorstep.

Here are the key aspects of the significance of antivirus software

- **Malware Detection and Prevention:**
  Antivirus software is designed to detect, prevent, and remove various types of malware, including viruses, worms, trojans, spyware, ransomware, and other malicious programs. It acts as a crucial defense mechanism against these threats.
- **Data Protection:**
  Antivirus programs help safeguard sensitive data stored on computer systems by preventing malware from corrupting or stealing information. This is especially important for personal and business data security.
- **Real-Time Scanning:**
  Many antivirus solutions provide real-time scanning capabilities, actively monitoring files, processes, and network activity to identify and block malware as soon as it is detected.
- **Email and Web Protection:**
  Antivirus programs often include features to scan email attachments and check websites for malicious content. This helps prevent users from inadvertently downloading malware from these sources.

MANAGEMENT OF HOST FIREWALL AND ANTI-VIRUS

MANAGEMENT OF HOST FIREWALL

Managing a host firewall involves configuring, monitoring, and maintaining the firewall settings on an individual device to ensure its security. Here is a step-by-step guide on managing a host firewall

- **Understand Security Requirements**
  Identify the specific security requirements of the device based on its role and function. Different devices may have different needs, and firewall configurations should align with these requirements.
- **Configure Default Deny Policy**
  Implement a default deny policy for both inbound and outbound traffic. This means that unless explicitly allowed, all traffic is denied. This helps minimize the attack surface and ensures a proactive security stance.
- **Define Firewall Rules**
  Create specific firewall rules to allow or block traffic based on protocols, ports, and IP addresses. Define rules for both incoming and outgoing traffic, considering the principle of least privilege.
- **Prioritize Firewall Rules**
  Prioritize firewall rules based on importance and relevance to the device's functionality. Critical services and applications should have higher priority, while unnecessary or risky traffic should be blocked.

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- **Regularly Review and Update Rules**
  Periodically review firewall rules to ensure they align with the device's current configuration and security requirements. Remove outdated rules and update configurations as needed.
- **Logging and Monitoring**
  Enable firewall logging to capture information about allowed and denied traffic. Regularly monitor firewall logs to identify potential security incidents, anomalies, or attempts at unauthorized access.
- **Update Firewall Software**
  Keep the host firewall software up to date by applying patches and updates. Regularly check for updates to ensure that the firewall benefits from the latest security features and bug fixes.
- **Educate End-Users**
  Educate end-users about the importance of the firewall and the potential risks associated with modifying firewall settings. Discourage users from disabling or bypassing the firewall without proper authorization.

MANAGEMENT OF ANTI-VIRUS

- **Choose reputable anti-virus software:** Opt for trusted brands with proven track records and good performance ratings.
- **Enable real-time scanning:** Ensure continuous monitoring of files, emails, and downloads for suspicious activity.
- **Schedule regular scans:** Configure automatic scans for full system checks at appropriate intervals, like daily or weekly.
- **Customize scanning settings:** Define specific file types or folders to be scanned thoroughly, while excluding trusted sources.
- **Quarantine management:** Configure how quarantined files are handled, including options for deletion, restoration, or further analysis.
- **Automatic updates:** Enable automatic updates for both the anti-virus program itself and its virus definitions to stay ahead of evolving threats.
- **System updates:** Keep your operating system and applications updated with the latest security patches to address vulnerabilities.
- **Performance monitoring:** Regularly monitor the anti-virus software's performance to ensure it doesn't significantly impact system resources.
- **Periodic reviews:** Regularly review your anti-virus settings and configurations to ensure they remain optimal for your needs.

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Wi-Fi security

Wi-Fi security is crucial for protecting your wireless network and the devices connected to it from unauthorized access, data breaches, and various cyber threats. Implementing robust security measures helps ensure the confidentiality and integrity of the data transmitted over the Wi-Fi network.

Here are essential Wi-Fi security measures

- **Use Strong Encryption**
  Enable WPA3 (Wi-Fi Protected Access 3) or, at a minimum, WPA2 encryption on your Wi-Fi router. Encryption scrambles the data transmitted between your device and the router, making it difficult for attackers to intercept and decipher.
- **Secure Network Name (SSID)**
  Change the default SSID (service set identifier, Wi-Fi network name) to a unique and non-obvious name. Avoid using easily guessable names that may disclose information about the router or network.
- **Strong and Unique Passwords**
  Set a strong and unique password for both your Wi-Fi network and the router's administration interface. Use a combination of uppercase and lowercase letters, numbers, and special characters. Avoid using default or easily guessable passwords.
- **Network Segmentation**
  Consider setting up a guest network for visitors. This isolates their devices from your main network, adding an extra layer of security and preventing potential threats from spreading to your primary devices.
- **Regularly Update Router Firmware**
  Keep your router's firmware up to date. Manufacturers release updates to address security vulnerabilities and enhance overall performance. Check for firmware updates periodically and apply them as needed.
- **MAC Address Filtering**
  Implement MAC address filtering on your router. This allows only specific devices with approved MAC addresses to connect to the network. However, note that MAC addresses can be spoofed.
- **Disable Remote Management**
  If possible, disable remote management of the router. This prevents attackers from accessing the router's settings from outside the local network.
- **Use a Virtual Private Network (VPN)**
  Consider using a VPN to encrypt your internet connection and add an extra layer of security, especially when connecting to public Wi-Fi networks.
- **Enable Network Logging**
  Enable logging on your router to keep track of network activity. Regularly review logs for any unusual or suspicious activities that may indicate unauthorized access.
- **Educate Users**
  Educate users about the importance of Wi-Fi security. Remind them not to share passwords casually and to be cautious when connecting to public Wi-Fi networks.

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